

ReLU integral probability metric and its applications

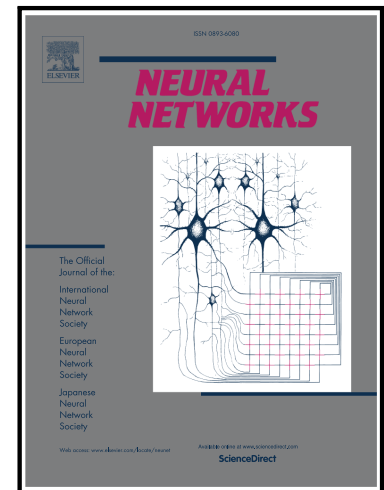
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Abstract

We propose a parametric integral probability metric (IPM) to measure the discrepancy between two probability measures. The proposed IPM leverages a specific parametric family of discriminators based on single-node neural networks with Rectified Linear Unit (ReLU) activation. The proposed IPM yields estimators with good convergence rates and can serve as a surrogate for other IPMs that use smooth nonparametric discriminator classes. We present an efficient algorithm for practical computation, offering a simple implementation and requiring fewer hyperparameters. Furthermore, we explore its applications in various tasks, such as covariate balancing for causal inference and fair representation learning. Across such applications, we demonstrate that algorithms with the proposed IPM provide strong theoretical guarantees, and empirical experiments show that they achieve comparable or even superior performance to other methods.

Keywords: integral probability metric, ReLU activation, convergence bound, covariance balancing, fair representation learning.

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1. Introduction

Measuring the discrepancy between two probability measures plays important roles in various AI applications involving deep neural networks. A representative example is probabilistic generative modeling, where methods such as generative adversarial networks (Goodfellow et al., 2014; Arjovsky et al., 2017), moment-matching generative models (Li et al., 2015) aim to learn a deep generative model by reducing the discrepancy between the generated and real data distributions. Another examples are domain adaptation (Long et al., 2018), fair representation learning (Madras et al., 2018) and causal inference (Wong and Chan, 2017; Kong et al., 2023) to name just a few.

Among various measures of such discrepancy, integral probability metrics (IPMs) (Müller, 1997) have received much attention for their conceptual clarity and attractive theoretical properties. Let $\mathcal{F}$ be a class of bounded measurable functions f from $\mathcal{X}$ to $\mathbb{R}$. For a given class $\mathcal{F}$, the IPM between two given probability measures $\mathcal{P}$ and $\mathcal{Q}$, denoted as $d_{\mathcal{F}}(\mathcal{P}, \mathcal{Q})$, is defined as the maximum average difference between $\mathcal{P}$ and $\mathcal{Q}$ on the class $\mathcal{F}$ over $\mathcal{X}$,

$$d_{\mathcal{F}}(\mathcal{P}, \mathcal{Q}) := \sup_{f \in \mathcal{F}} \left| \int f(d\mathcal{P} - d\mathcal{Q}) \right| = \sup_{f \in \mathcal{F}} \left| \mathbb{E}_{\mathbf{X}} f(\mathbf{X}) - \mathbb{E}_{\mathbf{Y}} f(\mathbf{Y}) \right|, \quad (1)$$

when $\mathbf{X} \sim \mathcal{P}$ and $\mathbf{Y} \sim \mathcal{Q}$. The functions in $\mathcal{F}$ are commonly referred to as *discriminators*, *critics*, or *adversaries*, depending on the application.

Different choices of the discriminator class $\mathcal{F}$ result in different IPMs. For example, the maximum mean discrepancy (MMD) (Gretton et al., 2012) is defined by selecting a reproducing kernel Hilbert space (RKHS) whereas the Wasserstein distance (Villani, 2009) uses the set of 1-Lipschitz functions as its discriminator class. The Hölder-IPM (Wang et al., 2023) and Sobolev-IPM (Mroueh et al., 2017; Wong and Chan, 2017) select the Hölder and Sobolev spaces, respectively, as their discriminator classes. In turn, different IPMs affect the final estimator differently. A larger discriminator class can identify more complex differences between two probability measures, and thus more complex patterns could be learned to yield less bias in the final estimator. In contrast, as usual, large discriminator classes would lead to over-fitting and thus higher variance. Moreover, numerical computation of IPMs with a larger discriminator class is more challenging. Thus, there is a need to find a discriminator class which is small but is able to identify complex patterns between two probability measures.

In contrast, parametric IPMs, that is, IPMs with parametric discriminator classes, are more stable, computationally lighter, and less prone to poor local minima. Kim et al. (2022a) considers a parametric discriminator class $\{\text{sig}(\boldsymbol{\theta}^\top \mathbf{z} + \mu) : \mu \in \mathbb{R}, \boldsymbol{\theta} \in \mathbb{R}^d\}$, where $\text{sig}(\cdot)$ is the sigmoid function (i.e. $\text{sig}(z) = (1 + \exp(-z))^{-1}$), whose corresponding IPM is called the sigmoid-IPM (SIPM). An interesting property of the SIPM is that it is theoretically equivalent to an IPM with a broader class (e.g., the set of all infinitely differentiable functions). Kim et al. (2022a) and Kong et al. (2023) empirically demonstrate that the SIPM competes well with existing nonparametric IPMs for fair representation learning and causal inference.

The aim of this paper is to propose a new parametric IPM, called the ReLU-IPM, where the discriminator class is given as $\{(\boldsymbol{\theta}^\top \mathbf{z} + \mu)_+ : \mu \in [-1, 1], \boldsymbol{\theta} \in \mathbb{S}^{d-1}\}$, where $(\cdot)_+ = \max(\cdot, 0)$ denotes the ReLU activation function and $\mathbb{S}^{d-1}$ is the unit sphere in $\mathbb{R}^d$. We provide an interesting theoretical relation between the ReLU-IPM and the Hölder-IPM in that the ReLU-IPM is an upper bound of the Hölder-IPM. Moreover, it is proved that the ReLU-IPM and Hölder-IPM are asymptotically equivalent when the Hölder discriminators are sufficiently smooth. Surprisingly, these results suggest that the ReLU-IPM has discriminative power comparable to that of nonparametric IPMs, even though its discriminator class has a simple parametric form.

An important advantage of ReLU-IPM compared to SIPM is that ReLU-IPM covers much larger classes of IPMs than SIPM does. That is, ReLU-IPM serves as an upper bound of IPMs with Hölder discriminators while SIPM does for IPMs with infinitely differentiable discriminators (Kim et al., 2022a). Note that the class of Hölder-smooth functions is much larger than the class of infinitely differentiable functions. See Remark 2 for additional discussions.

The Hölder-IPM has the advantage of controlling the size of the discriminator class by controlling the smoothness of the Hölder discriminators. In contrast, the discriminator classes of the Wasserstein distance, which include all of differentiable functions, would be too large while the discriminator class of the MMD, which consists of only very smooth functions (e.g., infinitely differentiable functions for the MMD with the Gaussian kernel), would be too small. However, computation of the Hölder-IPM is difficult since Hölder discriminators should be approximated by computable models such as deep neural networks (Wang et al., 2023). The choice of an appropriate architecture of DNNs for approximating Hölder discriminators would be hard, which makes the Hölder-IPM be underappreciated. Our theoretical results suggest that the ReLU-IPM can be used as a surrogate IPM for the Hölder-IPM.

For illustrating the usefulness of ReLU-IPM, in this paper, we consider two AI applications – causal inference and fair representation learning. For the causal problem, we use the ReLU-IPM in the covariate balancing (CB) estimator for the average treatment effect on the treated (ATT). In particular, when the true output regression is sufficiently smooth (Hölder-smooth), we prove that the CB estimator of ATT with the ReLU-IPM (ReLU-CB) achieves a parametric rate. By analyzing simulated data, we demonstrate that the ReLU-CB outperforms other competitors, including the SIPM. On the other hand, the objective of fair representation learning (FRL) is to train a neural network to produce representations that are useful for prediction but less informative about sensitive attributes (Zemel et al., 2013; Beutel et al., 2017; Madras et al., 2018). A common approach of FRL is to minimize the cross entropy penalized by a discrepancy between representation distributions across sensitive groups (Kim et al., 2022a; Deka and Sutherland, 2023). We consider using the ReLU-IPM for distributional discrepancy and prove that the fairness of the trained neural network is well controlled by the ReLU-IPM. Moreover, by analyzing several benchmark datasets, we demonstrate that the fair representation learning algorithm with the ReLU-IPM compares favorably with various baseline algorithms for learning fair neural networks.

The remainder of this paper is organized as follows: Section 2 provides a detailed review of IPMs, discussing their definitions, applications, and limitations in the context of machine learning. We introduce our proposed metrics, ReLU-IPM, in Section 3 and present its applications to the following tasks: causal inference and fair representation learning in Sections 4 and 5, respectively. Finally, Section 6 concludes the paper and suggests directions for future research.

Notations. Vectors and matrices are represented in boldface letters. Let $\mathbb{N} = \{1, 2, \dots\}$ be the set of natural numbers. For a given two sequence $\{a_n\}$ and $\{b_n\}$, the notation $a_n \lesssim b_n$ and $a_n \gtrsim b_n$ indicates that there exists a positive constant c such that for all n , $a_n \leq cb_n$ and $a_n \geq cb_n$, respectively. When $a_n \lesssim b_n$ and $a_n \gtrsim b_n$, we write $a_n \asymp b_n$. For $a \in \mathbb{R}$, let $\lfloor a \rfloor$ and $\lceil a \rceil$ denote the floor and ceiling functions, which round a to the next strictly smaller and larger integer, respectively. We denote $\mathbf{0}$ and $\mathbf{1}$ as the vectors of all zeros and all ones, respectively, in the corresponding dimensional space. For a given condition or event A , $\mathbb{I}(A)$ denotes the indicator function, where $\mathbb{I}(A) = 1$ if A is true and $\mathbb{I}(A) = 0$ if A is false.

2. Review of existing IPMs

In this section, we review popularly used IPMs.

Wasserstein distance. Given two distributions $\mathcal{P}$ and $\mathcal{Q}$, the p -Wasserstein distance is defined as

$$W_p(\mathcal{P}, \mathcal{Q}) := \left(\inf_{\pi \in \Pi(\mathcal{P}, \mathcal{Q})} \int_{\mathcal{X} \times \mathcal{X}} d(\mathbf{x}, \mathbf{y})^p d\pi(\mathbf{x}, \mathbf{y}) \right)^{1/p}, \quad (2)$$

where $d(\mathbf{x}, \mathbf{y})$ is a distance between two points $\mathbf{x}$ and $\mathbf{y}$, and $\Pi(\mathcal{P}, \mathcal{Q})$ is the set of all joint probability measures (couplings) $\pi(\mathbf{x}, \mathbf{y})$ with the marginals $\mathcal{P}$ and $\mathcal{Q}$. By the Kantorovich-Rubinstein duality (Kantorovich and Rubinshtein, 1958; Villani, 2009), the 1-Wasserstein distance can be expressed as the IPM with a discriminator class containing all bounded 1-Lipschitz functions. That is, the 1-Wasserstein distance can be redefined as $d_{\mathcal{F}_L}(\mathcal{P}, \mathcal{Q}) := \sup_{f \in \mathcal{F}_L} |\mathbb{E}_{\mathbf{X}} f(\mathbf{X}) - \mathbb{E}_{\mathbf{Y}} f(\mathbf{Y})|$, where $\mathcal{F}_L := \{f : \|f\|_L \leq 1\}^1$. Originally derived from the theory of optimal transport, it represents the minimal transportation cost required to move one distribution into another, considering both the distance and the amount of mass that must be moved. Thus, it captures geometric differences between two distributions.

Recently, the Wasserstein distance has gained popularity in the domain of image generation (Arjovsky et al., 2017; Nadjahi et al., 2019), where it helps to improve the stability and convergence of the training process by addressing issues such as mode collapse. Furthermore, it has been used in various tasks, including domain adaptation (Shen et al., 2018; Xu et al., 2019), object detection (Xu et al., 2018; Garg et al., 2020; Yang et al., 2021), image retrieval (Rubner et al., 2000), reinforcement learning (Bellemare et al., 2017), distributionally robust optimization (Blanchet et al., 2019), optimal transport problems (Villani, 2009; Cuturi, 2013) statistical testing (del Barrio et al., 1999; Hallin et al., 2021), and representation learning (Ozair et al., 2019). It can be computed using an algorithm that employs a DNN to approximate 1-Lipschitz functions (Gulrajani et al., 2017; Arjovsky et al., 2017), but approximating 1-Lipschitz functions is challenging because choosing an appropriate architecture is difficult and optimization is often unstable.

¹ $\|f\|_L := \sup_{\mathbf{x} \neq \mathbf{y}} \frac{|f(\mathbf{x}) - f(\mathbf{y})|}{\|\mathbf{x} - \mathbf{y}\|}$ is the Lipschitz semi-norm of a real-valued function $f : \mathcal{X} \rightarrow \mathbb{R}$, where $\|\cdot\|$ is a certain norm defined on $\mathcal{X}$ (i.e., $\|f\|_L$ is the Lipschitz constant of the function f).

Maximum mean discrepancy (MMD). The discriminator class for MMD is $\mathcal{H}_k := \{f : \|f\|_{\mathcal{H}_k} \leq 1\}$, where $\mathcal{H}_k$ is the reproducing kernel Hilbert space associated with a given kernel k . That is, the MMD is defined as $d_{\mathcal{H}_k}(\mathcal{P}, \mathcal{Q}) := \sup_{f \in \mathcal{H}_k} |\mathbb{E}_{\mathbf{X}} f(\mathbf{X}) - \mathbb{E}_{\mathbf{Y}} f(\mathbf{Y})|$. An important advantage of the MMD is that it can be computed in a closed form. This leads to the well-known expression in terms of a given kernel function as

$$d_{\mathcal{H}_k}^2(\mathcal{P}, \mathcal{Q}) = \mathbb{E}k(\mathbf{X}, \mathbf{X}') - 2\mathbb{E}k(\mathbf{X}, \mathbf{Y}) + \mathbb{E}k(\mathbf{Y}, \mathbf{Y}').$$

The MMD is used in statistical testing (Gretton et al., 2012), generative modeling (Li et al., 2015; Dziugaite et al., 2015; Li et al., 2017; Galashov et al., 2025; De Bortoli et al., 2025), causal inference (Wong and Chan, 2017; Kong et al., 2023), and domain adaptation (Zhang et al., 2012; Yan et al., 2017). The MMD is computationally efficient, easy to implement, and generally performs well for high-dimensional data. However, its performance depends strongly on the choice of kernel function. The Gaussian kernel is popularly used (Vert et al., 2004; Gretton et al., 2012; Tolstikhin et al., 2016; Garreau et al., 2018; Li and Yuan, 2024), but its associated RKHS, which consists only of infinitely differentiable functions (Micchelli et al., 2006; Kroese et al., 2019), would be too small and selecting an appropriate bandwidth in practice could be difficult.

Hölder-IPM. The Hölder-IPM (Wang et al., 2023; Tang and Yang, 2023) uses the Hölder function class, $\mathcal{H}_d^\beta := \{f : \|f\|_{\mathcal{H}_d^\beta} \leq 1\}$, for the discriminator class, where the Hölder norm is defined as

$$\|f\|_{\mathcal{H}_d^\beta} = \sum_{\|\alpha\|_1 = \lfloor \beta \rfloor} \max_{x, y \in \Omega, x \neq y} \frac{|f^{(\alpha)}(x) - f^{(\alpha)}(y)|}{\|x - y\|_2^{\beta - \lfloor \beta \rfloor}} + \sum_{\|\alpha\|_1 \leq \lfloor \beta \rfloor} \max_{x \in \Omega} |f^{(\alpha)}(x)|,$$

where $f^{(\alpha)} = \frac{\partial^{\|\alpha\|_1} f}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \dots \partial x_d^{\alpha_d}}$ for $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_d)^\top$. That is, the Hölder-IPM is defined as $d_{\mathcal{H}_d^\beta}(\mathcal{P}, \mathcal{Q}) := \sup_{f \in \mathcal{H}_d^\beta} |\mathbb{E}_{\mathbf{X}} f(\mathbf{X}) - \mathbb{E}_{\mathbf{Y}} f(\mathbf{Y})|$. Wang et al. (2023) considers the Hölder-IPM for statistical testing problems, and states that practical applications of the Hölder-IPM become feasible through the approximation of Hölder smooth functions by the use of neural networks because it is well known that neural networks are good at approximating Hölder smooth functions (Ohn and Kim, 2019; Yang and Zhou, 2025). However, the choice of neural network architectures is difficult.

Sigmoid-IPM (SIPM). The SIPM introduced by Kim et al. (2022a) is an IPM with a parametric discriminator class, where the discriminator class $\mathcal{F}_{\text{sig}}$ is given as

$$\mathcal{F}_{\text{sig}} := \{f : f(\mathbf{z}) = \text{sig}(\boldsymbol{\theta}^\top \mathbf{z} + \mu), \boldsymbol{\theta} \in \mathbb{R}^d, \mu \in \mathbb{R}\}.$$

That is, the SIPM is defined as $d_{\mathcal{F}_{\text{sig}}}(\mathcal{P}, \mathcal{Q}) := \sup_{f \in \mathcal{F}_{\text{sig}}} |\mathbb{E}_{\mathbf{X}} f(\mathbf{X}) - \mathbb{E}_{\mathbf{Y}} f(\mathbf{Y})|$. The development and application of the SIPM are well-studied in fair representation learning (FRL) (Kim et al., 2022a) and causal inference (Kong et al., 2023). An interesting theoretical property of the SIPM obtained by Kim et al. (2022a) is that the SIPM is equivalent to the IPM whose discriminator class is the set of all infinitely differentiable functions.

3. ReLU-IPM: New parametric IPM

In this section, we propose a new IPM with a parametric discriminator class $\mathcal{F}$ that is simple but effective. For $\mathcal{F}$ to be a suitable choice, it must be computationally efficient while maintaining sufficient flexibility to capture a broad range of discriminators larger than $\mathcal{F}$.

3.1. Definition

We consider the discriminator class $\mathcal{F}_{\text{ReLU}}$ defined on the unit ball $\mathbb{B}^d = \{\mathbf{z} \in \mathbb{R}^d : \|\mathbf{z}\|_2 \leq 1\}$ of $\mathbb{R}^d$ given as

$$\mathcal{F}_{\text{ReLU}} := \{f : \mathbb{B}^d \rightarrow \mathbb{R} \mid f(\mathbf{z}) = (\boldsymbol{\theta}^\top \mathbf{z} + \mu)_+, \mu \in [-1, 1], \boldsymbol{\theta} \in \mathbb{S}^{d-1}\}, \quad (3)$$

where $\mathbb{S}^{d-1} = \{\mathbf{z} \in \mathbb{R}^d : \|\mathbf{z}\|_2 = 1\}$ is the unit sphere in $\mathbb{R}^d$. Then we define the ReLU-IPM as

$$d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}) = \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \mathbb{E}_{\mathbf{X} \sim \mathcal{P}} f(\mathbf{X}) - \mathbb{E}_{\mathbf{Y} \sim \mathcal{Q}} f(\mathbf{Y}) \right|,$$

for the two probability measures $\mathcal{P}$ and $\mathcal{Q}$ defined on $\mathbb{B}^d$.

By replacing the population expectations with the empirical counterparts based on the samples $\mathbf{X}^n = \{\mathbf{X}_1, \dots, \mathbf{X}_n\}$ and $\mathbf{Y}^m = \{\mathbf{Y}_1, \dots, \mathbf{Y}_m\}$, it is natural to define the empirical estimate of the ReLU-IPM as

$$d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_n, \hat{\mathcal{Q}}_m) := \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{X}_i) - \frac{1}{m} \sum_{i=1}^m f(\mathbf{Y}_i) \right|, \quad (4)$$

where $\hat{\mathcal{P}}_n(\cdot) = \frac{1}{n} \sum_{i=1}^n \delta_{\mathbf{X}_i}(\cdot)$ and $\hat{\mathcal{Q}}_m(\cdot) = \frac{1}{m} \sum_{i=1}^m \delta_{\mathbf{Y}_i}(\cdot)$. Here, δ_x denotes the Dirac measure on x .

3.2. Theoretical Studies of ReLU-IPM

The first theoretical property is that the ReLU-IPM can perfectly distinguish any two probability measures, as presented in the following proposition.

Proposition 1. $d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}) = 0$ if and only if $\mathcal{P} \equiv \mathcal{Q}$ for two probability measures $\mathcal{P}$ and $\mathcal{Q}$.

This proposition implies that $d_{\mathcal{F}_{\text{ReLU}}}$ can distinguish between $\mathcal{P}$ and $\mathcal{Q}$, even though it is induced by a parametric family. The proof of Proposition 1 is deferred to Appendix B.1.

Remark 1. *It would not be very surprising that a parametric IPM perfectly distinguishes any two probability measures. The characteristic function distance (CFD), which is the IPM with discriminator class $\mathcal{F}_{\text{cf}} = e^{it^\top \mathbf{x}} : \mathbf{t} \in \mathbb{R}^d$, is introduced by Ansari et al. (2020). It is well known that $d_{\mathcal{F}_{\text{cf}}}(\mathcal{P}, \mathcal{Q}) = 0$ if and only if $\mathcal{P} = \mathcal{Q}$. However, McCullagh (1994) notices that $d_{\mathcal{F}_{\text{cf}}}$ would not be a useful metric between two probability measures.*

The following theorem is one of the main results of this paper, which describes the relationship between the ReLU-IPM and Hölder-IPM. In Theorem 2 below, we prove that the Hölder-IPM is upper bounded by the ReLU-IPM.

Theorem 2 (ReLU-IPM bounds Hölder-IPM). *Let $d \in \mathbb{N}$ and $\beta > 0$ be given. For some positive constant c that depends on d and β , we have*

(i) if $\beta < \frac{d+3}{2}$,

$$d_{\mathcal{H}_d^\beta}(\mathcal{P}, \mathcal{Q}) \leq c \cdot d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q})^{\frac{2\beta}{d+3}},$$

(ii) and if $\beta \geq \frac{d+3}{2}$,

$$d_{\mathcal{H}_d^\beta}(\mathcal{P}, \mathcal{Q}) \leq c \cdot d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}),$$

for any two probability measures $\mathcal{P}$ and $\mathcal{Q}$ in $\mathbb{B}^d$.

This result implies that the ReLU-IPM can be used as a surrogate for the Hölder-IPM. That is, we can reduce the Hölder-IPM by reducing the ReLU-IPM. The proof of Theorem 2 is provided in Appendix B.2. By the definition of IPMs, it is obvious that an IPM with a larger discriminator class has a larger value than IPMs with smaller discriminator classes. In fact, from $\mathcal{F}_{\text{ReLU}} \subset \mathcal{H}_d^1$ we have $d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}) \leq d_{\mathcal{H}_d^1}(\mathcal{P}, \mathcal{Q})$ for any $\beta \leq 1$. In contrast, Theorem 2 provides an opposite result. That is, a small value of the

ReLU-IPM guarantees a small value of the Hölder-IPM. It is remarkable that the IPM utilizing the parametric class $\mathcal{F}_{\text{ReLU}}$, which is significantly smaller than the nonparametric class $\mathcal{H}_d^\beta$, has such a property.

Remark 2. *The SIPM also exhibits a similar property, allowing it to serve as a surrogate of IPM that employs a set of infinitely many differentiable functions. However, the size of the set of infinitely differentiable functions is comparable to that of a parametric class, as one can still achieve parametric convergence rates over this class.*

Since the ReLU-IPM uses a parametric discriminator class, it has many desirable statistical properties. One of such desirable properties is a fast convergence of the empirical ReLU-IPM to its population version as stated in the following theorem, whose proof is given in Appendix B.3.

Theorem 3 (Rate of convergence of the empirical ReLU-IPM). *Let $\mathcal{P}, \mathcal{Q}, \hat{\mathcal{P}}_n, \hat{\mathcal{Q}}_n$ be defined on $\mathbb{B}^d$. Then, for a constant c ,*

$$\left| d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}) - d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_n, \hat{\mathcal{Q}}_n) \right| \leq \frac{c}{\sqrt{n}} + \epsilon$$

with probability at least $1 - 2 \exp\left(-\frac{\epsilon^2 n}{16}\right)$. Furthermore, $d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_n, \hat{\mathcal{Q}}_n)$ converges almost surely to $d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q})$, i.e.,

$$\left| d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}) - d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_n, \hat{\mathcal{Q}}_n) \right| \xrightarrow{a.s.} 0 \quad \text{as } n \rightarrow \infty.$$

Note that the convergence rate is parametric and not depending on the dimension d of data, which is a highly desirable property. In contrast, the convergence rate of the empirical Wasserstein distance is $\mathcal{O}(n^{-\frac{1}{2}} \log n)$ when $d = 1$ but $\mathcal{O}(n^{-\frac{1}{d+1}})$ when $d \geq 2$ (Sriperumbudur et al., 2009, 2012). The SIPM can also achieve parametric convergence rates when $d = 1$, but it is challenging to establish such rates in higher dimensions.

3.3. An algorithm for computing the ReLU-IPM

For the computation of the ReLU-IPM, we use the gradient projection method. That is, given the current values of parameters μ^{old} and θ^{old} , we take the gradients of μ^{old} and θ^{old} , and update the parameters to μ^{new} and θ^{new} . We then keep μ^{new} fixed but normalize θ^{new} to have the unit norm.

The gradient projection method often gets stuck in a local optimum. To overcome this difficulty, we randomly select K initial parameters $\{\mathbf{v}_i\}_{i=1}^K := \{(\mu_i, \boldsymbol{\theta}_i)\}_{i=1}^K$, and update the parameters $\mathbf{v}_i$ s simultaneously. Finally, we select the parameter which has the highest the ReLU-IPM value. A pseudo-code for computing the ReLU-IPM is presented in Algorithm 1.

Note that updating $\mathbf{v}_i$ s can be done by a parallel computation. Since $f_{\mathbf{v}_1}, \dots, f_{\mathbf{v}_K} \in \mathcal{F}_{\text{ReLU}}$ can be represented via a single affine map $\mathbf{X} \mapsto (\Theta \mathbf{X} + \boldsymbol{\mu})_+ \in \mathbb{R}^K$, where $\Theta = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_K)^\top$ and $\boldsymbol{\mu} = (\mu_1, \dots, \mu_K)^\top$, backpropagation can be carried out in parallel in the python package without any special program, and thus its computing time does not increase much with respect to K . See Appendix E for empirical evidences.

Algorithm 1 Calculation of ReLU-IPM

- 1: Initialize $\mathbf{v} = (\mathbf{v}_1, \dots, \mathbf{v}_K)$
 - 2: **for** $t = 1, 2, \dots, n_{\text{epoch}}$ **do**
 - 3: Calculate $L(\mathbf{v}) = \sum_{k=1}^K \left\{ \frac{1}{n} \sum_{i=1}^n f_{\mathbf{v}_k}(\mathbf{X}_i) - \frac{1}{m} \sum_{i=1}^m f_{\mathbf{v}_k}(\mathbf{Y}_i) \right\}^2$
 - 4: $\mathbf{v} \leftarrow \mathbf{v} + \eta \nabla_{\mathbf{v}} L(\mathbf{v})$
 - 5: **end for**
 - 6: **return** $\max_{1 \leq k \leq K} \left| \frac{1}{n} \sum_{i=1}^n f_{\mathbf{v}_k}(\mathbf{X}_i) - \frac{1}{m} \sum_{i=1}^m f_{\mathbf{v}_k}(\mathbf{Y}_i) \right|$
-

4. Application 1: Estimation of causal effects using ReLU-IPM

The estimation of causal effects from observational data has become an important research topic in AI (Yao et al., 2021). Most AI models are trained on passively collected data, but decision-making often requires estimating how outcomes would change under alternative actions. Thus, developing methods for AI models to learn causal relations from observational data has become an important topic, not only for tabular data but also for image and language data (Feder et al., 2022; Jerzak et al., 2023; Dhawan et al., 2024).

A fundamental challenge in observational causal inference is that the covariate distributions of the treated and control groups may differ substantially. To adjust such systematic biases, various methods have been developed including regression adjustment (Hill, 2011), matching (Stuart, 2010), and weighting (Imai and Ratkovic, 2014). The common goal of these methods is to make the covariate distributions of the control and treated groups be (asymptotically) balanced.

A fundamental technique for achieving covariate balance is inverse probability weighting (IPW) (Hirano et al., 2003). In IPW, each sample is assigned

a weight that is reciprocally proportional to its propensity score, defined as the conditional probability of treatment assignment given the covariates. One issue in using the IPW is that its finite sample property would be inferior because accurately estimating the propensity score does not automatically mean the accurate estimation of its inverse (Li et al., 2018).

Directly estimating the weights for covariate balancing has been received much attention. Hainmueller (2012) and Imai and Ratkovic (2014) propose to find the weights matching the sample moments of the covariates. These methods, however, are asymptotically valid only when the output regression model is linear in covariates. To resolve this problem, estimating the weights by use of IPMs has also been considered. Wong and Chan (2017) uses the MMD, and Kong et al. (2023) provides sufficient conditions on general IPMs for the consistency of the causal effect estimation.

In this section, we apply the ReLU-IPM to estimate covariate-balancing weights. We establish the consistency of the resulting causal effect estimator under the assumption that the true outcome regression function is Hölder smooth, and derive its convergence rate. By analyzing simulated data, we empirically demonstrate that the ReLU-IPM outperforms other IPMs, including the SIPM and MMD, for estimation of causal effect.

4.1. Estimation of ATT

In this study, we focus on estimating the average treatment effect on the treated (ATT), which represents the causal effect of the treatment in terms of how much the treated units benefit from it, viewed retrospectively, in the context of a binary treatment problem. Let $\mathbf{X} \in \mathcal{X} \subset \mathbb{R}^d$ be a random vector of covariates whose distribution is denoted by $\mathcal{P}$. Let $\pi(\cdot) = \mathbb{P}(T = 1 | \cdot)$ be the propensity score. We assume that there exists an $\eta > 0$ such that $\eta < \pi(\cdot) < 1 - \eta$. Also, let $T \in \{0, 1\}$ denote the binary treatment indicator, which is drawn from $\text{Ber}(\pi(\mathbf{x}))$ conditioned on $\mathbf{X} = \mathbf{x}$, and $Y := TY(1) + (1 - T)Y(0)$ be an observed outcome where $Y(0)$ and $Y(1)$ are referred to as the potential outcomes under control and treatment, respectively.

Suppose that we observe n independent copies $\mathcal{D}^{(n)} := \{(\mathbf{X}_i, T_i, Y_i)\}_{i=1}^n$ of $(\mathbf{X}, T, Y)$. Let n_0 and n_1 be the sizes of the control ($T=0$) and treatment ($T=1$) groups, respectively. Our goal is to estimate $\text{ATT} := \mathbb{E}(Y(1) - Y(0) | T=1)$ using $\mathcal{D}^{(n)}$. We consider the weighted estimator of ATT,

$$\widehat{\text{ATT}}^{\mathbf{w}} = \sum_{i:T_i=1} \frac{1}{n_1} Y_i - \sum_{i:T_i=0} w_i Y_i,$$

which is parametrized by $\mathbf{w}$, where $\mathbf{w} = (w_1, \dots, w_n)^\top$ is a nonnegative weight vector. We use $\mathbf{w}$ such that $\hat{\mathcal{P}}_1(\cdot) = \frac{1}{n_1} \sum_{i:T_i=1} \delta_{\mathbf{X}_i}(\cdot)$ and $\hat{\mathcal{P}}_0^{\mathbf{w}}(\cdot) = \sum_{i:T_i=0} w_i \delta_{\mathbf{X}_i}(\cdot)$ are as similar as possible. We propose to measure the similarity between $\hat{\mathcal{P}}_1(\cdot)$ and $\hat{\mathcal{P}}_0^{\mathbf{w}}(\cdot)$ using the ReLU-IPM. That is, we use

$$\hat{\mathbf{w}} = \underset{\mathbf{w} \in \mathcal{W}^+}{\operatorname{argmin}} d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_0^{\mathbf{w}}, \hat{\mathcal{P}}_1), \quad (5)$$

where the set of weight vectors $\mathcal{W}^+$ is defined as

$$\left\{ \mathbf{w} = (w_1, \dots, w_n)^\top \in [0, 1]^n : \max_{i \in \{1, \dots, n\}} w_i \leq \frac{K}{n_0}, \sum_{i:T_i=0} w_i = 1, \sum_{i:T_i=1} w_i = 0 \right\},$$

for a positive sufficiently large number K such that $1/\eta^2 \leq K$. We refer to $\widehat{\text{ATT}}^{\hat{\mathbf{w}}}$ as the *ReLU-CB estimator*.

Let $m_t(\cdot) = \mathbb{E}(Y(t)|\mathbf{X} = \cdot)$ for $t \in \{0, 1\}$. Theorem 4.2 of Kong et al. (2023) can be used to prove the consistency of the ReLU-CB estimator under regularity conditions, when m_0 is smooth. However, we can do more due to nice theoretical properties of the ReLU-IPM. That is, we can derive the convergence rate of the ReLU-CB estimator whose results are presented in Theorem 4 below.

Theorem 4. *Suppose that $m_0(\cdot)$ belongs to the Hölder class $\mathcal{H}_d^\beta$. Then,*

(i) *if $\beta < \frac{d+3}{2}$, we have*

$$\widehat{\text{ATT}}^{\hat{\mathbf{w}}} - \text{ATT} = \mathcal{O}_p\left(n^{-\frac{\beta}{d+3}}\right),$$

(ii) *and if $\beta \geq \frac{d+3}{2}$, we have*

$$\widehat{\text{ATT}}^{\hat{\mathbf{w}}} - \text{ATT} = \mathcal{O}_p\left(n^{-\frac{1}{2}}\right).$$

In particular, it is proved that the ReLU-CB estimator has a parametric convergence rate when the true outcome regression model is sufficiently smooth. The proof of Theorem 4 is deferred to Appendix B.4.

4.2. Simulation study

4.2.1. Settings

Simulation setup. We compare the proposed estimator with existing methods under two nonlinear simulation scenarios adapted from Kang and Schafer (2007) and Orihara et al. (2025). Both scenarios consider nonlinear treatment assignment and outcome models under two overlap levels ($\tau = 1$ and $\tau = 10$). The resulting settings are denoted by A1 and A2 for Scenario A and B1 and B2 for Scenario B, where the indices 1 and 2 correspond to $\tau = 1$ and $\tau = 10$, respectively. Scenario A considers homogeneous treatment effects, whereas Scenario B considers heterogeneous treatment effects. Complete descriptions of the data-generating mechanisms are given in Appendix C.1.

Baselines. As baselines, we consider covariate balancing methods, including (i) the stabilized IPW (SIPW) with the linear logistic regression (GLM) (Lunceford and Davidian, 2004), (ii) the SIPW with the boosting algorithm (Boost) (Lee et al., 2010), (iii) the SIPW with the covariate balancing propensity score (CBPS) with the linear logistic regression (Imai and Ratkovic, 2014), and (iv) the entropy balancing (EB) method (Hainmueller, 2012). For CBPS and EB, we match the first moments of $\mathbf{X}$. For baselines using IPMs, we consider five IPMs, including (iv) Wasserstein distance (Wass), (v) sigmoid-IPM (SIPM), (vi) MMD with the RBF kernel (RBF), (vii) MMD with the Sobolev kernel (Sob) of Wong and Chan (2017), and (viii) Hölder-IPM using DNN discriminators with L -many layers for $L = 1, 2$ (Höl-1, Höl-2). Details of the baselines are provided in Appendix C.2.

4.2.2. Results

Table 1 shows the results for the four simulation settings based on 1000 simulated datasets. In most cases, the ReLU-CB outperforms the other estimators with large margins in terms of both bias and RMSE. The Höl-1 and Höl-2 tend to have larger RMSEs than ReLU-CB, which may be partly due to overfitting. Moreover, their RMSEs do not consistently decrease with increasing sample size. The GLM and Boost, which rely on the inverse of the estimated propensity score, perform substantially worse than covariate balancing methods, confirming that accurate estimation of the propensity score does not necessarily guarantee accurate estimation of its inverse. The relatively small bias of the ReLU-CB is noteworthy, given that the ReLU-IPM is parametric while the true outcome regression model is highly nonlinear.

Table 1: **Simulation results.** We generate 1000 simulated datasets from the four models from A1 to B2, with two sample sizes, and report the biases and RMSEs. For each row, the best result is highlighted in bold, while the second best result is underlined.

Model	Meas.	n	GLM	Boost	CBPS	EB	Wass	SIPM	RBF	Sob	Höl-1	Höl-2	ReLU-CB
A1	Bias	200	-7.820	-8.375	-4.745	-4.806	-3.934	-3.567	<u>-3.162</u>	-3.255	-3.358	-4.935	-2.448
		1000	-7.410	-6.489	-4.496	-4.494	-3.607	-2.757	-2.844	-2.700	<u>-2.347</u>	-4.787	-1.819
	RMSE	200	8.719	9.204	5.354	5.395	4.619	4.611	4.356	<u>4.285</u>	7.995	9.255	3.952
		1000	7.622	6.665	4.620	4.618	3.760	3.009	3.020	<u>2.861</u>	3.905	9.607	2.080
A2	Bias	200	-7.753	-8.306	-4.671	-4.732	-3.882	-3.505	-3.147	-3.205	-3.492	<u>-2.633</u>	-2.445
		1000	-7.452	-6.525	-4.540	-4.538	-3.677	-2.824	-2.878	-2.722	<u>-2.400</u>	-2.690	-1.872
	RMSE	200	8.813	9.289	5.575	5.615	4.991	5.177	4.942	4.841	8.173	4.576	<u>4.733</u>
		1000	7.707	6.753	4.730	4.728	3.926	3.221	3.207	<u>3.034</u>	4.087	4.677	2.376
B1	Bias	200	0.222	0.134	0.192	0.193	0.145	0.076	0.044	0.011	0.079	0.100	0.037
		1000	0.226	0.093	0.201	0.201	0.171	0.139	0.013	-0.004	0.018	0.114	<u>0.005</u>
	RMSE	200	0.310	0.238	0.281	0.283	0.256	<u>0.239</u>	<u>0.254</u>	<u>0.253</u>	0.479	0.514	0.237
		1000	0.243	0.124	0.220	0.221	0.195	0.168	<u>0.127</u>	<u>0.123</u>	0.167	0.218	0.112
B2	Bias	200	0.317	0.256	0.274	0.276	0.231	0.145	<u>0.090</u>	0.087	0.241	0.228	0.120
		1000	0.314	0.235	0.283	0.285	0.260	0.130	0.066	<u>0.055</u>	0.150	0.138	0.048
	RMSE	200	0.410	0.332	0.354	0.358	0.326	<u>0.276</u>	0.275	0.281	0.779	0.733	0.287
		1000	0.331	0.253	0.298	0.301	0.277	0.197	0.196	<u>0.183</u>	0.492	0.489	0.158

This may be explained by the fact that the ReLU-IPM behaves similarly to the Hölder-IPM, as shown in Theorem 2.

4.3. Analysis on an additional benchmark dataset

We further evaluate the ReLU-CB estimator using the ACIC 2016 benchmark datasets (Dorie et al., 2019). These datasets were developed for the 2016 Atlantic Causal Inference Conference competition and combine real covariates with simulated treatment assignments and outcomes. Across the 77 simulation scenarios, treatment and outcome data were generated 100 times using real-world data consisting of 4,802 individuals and 58 covariates. From these 77 scenarios, we selected the first five and the last five (ten scenarios in total) and analyzed all 100 simulated datasets for each scenario.

The results in Table 2 indicate that the ReLU-CB demonstrates superior performance compared to other methods in terms of both performance measures. While the GLM yields better results in terms of Bias, the Wass and RBF exhibit better performance in terms of RMSE. Interestingly, the Höl-1 and Höl-2 shows smaller Bias values but larger RMSE values.

Table 2: **ACIC 2016 datasets.** For each row, the best result is highlighted in bold, while the next best result is underlined. We omit the columns of EB and Boost because solution of them often does not converge.

Dataset	Measure	GLM	CBPS	Wass	SIPM	RBF	Sob	Höl-1	Höl-2	ReLU-CB
1	Bias	0.022	0.091	0.052	0.071	0.093	0.153	-0.071	-0.489	<u>0.036</u>
	RMSE	0.314	0.237	<u>0.159</u>	0.212	0.160	0.249	0.941	0.956	0.122
2	Bias	-0.318	0.029	<u>0.024</u>	0.045	0.028	0.048	-0.020	-0.139	0.028
	RMSE	2.687	0.452	0.337	0.364	0.256	0.314	0.832	0.631	<u>0.269</u>
3	Bias	-0.017	0.134	0.076	0.100	0.135	0.210	0.099	-0.619	<u>0.050</u>
	RMSE	0.588	0.287	<u>0.164</u>	0.237	0.188	0.288	1.068	0.936	0.127
4	Bias	0.356	0.350	0.260	0.333	0.208	0.308	0.225	-0.123	<u>0.177</u>
	RMSE	0.789	0.766	0.533	0.748	<u>0.424</u>	0.599	1.012	0.714	0.375
5	Bias	-0.076	0.128	0.085	0.137	0.115	0.163	0.094	-0.183	<u>0.077</u>
	RMSE	2.210	0.840	0.543	0.404	<u>0.369</u>	0.440	1.224	0.946	0.299
6	Bias	0.432	0.467	0.372	0.403	0.406	0.449	<u>0.277</u>	-0.249	0.337
	RMSE	0.712	0.739	<u>0.608</u>	0.700	0.700	0.698	1.006	0.739	0.537
7	Bias	0.100	0.102	0.094	0.106	0.123	0.111	0.009	-0.196	<u>0.088</u>
	RMSE	0.368	0.354	<u>0.317</u>	0.365	<u>0.383</u>	0.327	1.042	0.679	0.269
8	Bias	0.248	0.273	0.224	0.249	0.245	0.249	0.137	-0.188	<u>0.200</u>
	RMSE	0.625	0.648	<u>0.545</u>	0.599	0.606	0.609	1.080	0.664	0.489
9	Bias	0.294	0.304	<u>0.243</u>	0.302	0.319	0.300	0.246	-0.203	<u>0.211</u>
	RMSE	0.534	0.525	<u>0.432</u>	0.505	0.516	0.490	0.866	0.636	0.383
10	Bias	0.420	0.421	0.341	0.390	0.410	0.392	0.405	-0.179	<u>0.293</u>
	RMSE	0.783	0.812	<u>0.641</u>	0.736	0.742	0.723	1.044	0.746	0.550

5. Application 2: Fair Representation Learning with ReLU-IPM

The notion of group fairness is a fundamental principle in algorithmic fairness, which aims to eliminate biases between sensitive groups (e.g. man vs woman, white vs black, etc.) divided by a given sensitive attribute (e.g. gender, race, etc.). The most popular criterion for group fairness is Demographic Parity (DP) which focuses on statistical disparities in decisions/outcomes across sensitive groups (Calders et al., 2009; Feldman et al., 2015; Agarwal et al., 2019; Chzheng et al., 2020; Jiang et al., 2020). Subsequent fairness metrics, including Equal Opportunity and Equalized Odds, suggest that fair decisions should be conditioned on both the label and the sensitive attribute (Hardt et al., 2016). Various algorithms have been proposed to achieve specified levels of group fairness with respect to one of these criteria (Zafar et al., 2017; Donini et al., 2018; Agarwal et al., 2018; Madras et al., 2018; Zafar et al., 2019; Chuang and Mroueh, 2021; Kim et al., 2022a).

Fair Representation Learning (FRL) aims at searching a fair representation

space (Zemel et al., 2013) in the sense that the distributions of the encoded representation vector for each sensitive group are similar. Since any prediction model built on a fair representation is automatically group-fair, FRL can be viewed as a tool for transforming the original data into a representation which can be used for downstream fair predictions. Numerous algorithms for FRL have been proposed, including (Louizos et al., 2015; Edwards and Storkey, 2016; Xie et al., 2017; Madras et al., 2018; Quadrianto et al., 2019; Gupta et al., 2021; Zeng et al., 2021; Kim et al., 2022a; Guo et al., 2022; Deka and Sutherland, 2023; Kong et al., 2025), to name just a few.

A popular learning strategy for FRL is the adversarial learning (Edwards and Storkey, 2016; Madras et al., 2018), where an adversary tries to predict a sensitive attribute on the representation space while the representation space is trained to prevent it. Kim et al. (2022a) apply the SIPM to the adversarial learning where the discriminator plays a role of an adversary and prove that any prediction model constructed on the learned fair representation space is also fair as long as the prediction head is infinitely differentiable. In addition, it is empirically demonstrated that FRL with the SIPM attains better fairness-accuracy trade-offs compared to existing baselines, including LAFTR (Madras et al., 2018).

In this section, we propose an FRL algorithm based on the ReLU-IPM. We prove that any prediction model on the learned fair representation space by the ReLU-IPM is guaranteed to be group-fair as long as the prediction model is Hölder continuous.

5.1. Preliminaries for Fair Representation Learning

We consider a binary sensitive attribute. Let $\mathbf{X} \in \mathcal{X} \subset \mathbb{R}^d$, $Y \in \mathcal{Y}$, and $S \in \{0, 1\}$ represent the d -dimensional input vector, the output variable, and the binary sensitive attribute. We write $\mathcal{P}$ and $\mathcal{P}_s$ as the joint distribution of $(\mathbf{X}, Y, S)$ and the conditional distribution of $\mathbf{X}|S = s$, respectively. Similarly, let $\mathbb{E}$ and $\mathbb{E}_s$ be the corresponding expectation operators.

A mapping $h(\mathbf{X}, S)$ from $\mathcal{X} \times \{0, 1\}$ to a representation space $\mathcal{Z}$ is called a fair representation if $h(\mathbf{X}, S)|S=0 \stackrel{d}{\approx} h(\mathbf{X}, S)|S=1$. For any prediction head $g: \mathcal{Z} \rightarrow \mathcal{Y}$, we then expect $g \circ h(\mathbf{X}, S)|S=0 \stackrel{d}{\approx} g \circ h(\mathbf{X}, S)|S=1$ as long as h is fair. Thus, FRL can be viewed as learning a transformation of the original data that enables downstream group-fair predictions.

For a given representation h , let $\mathcal{P}_s^h$ be the conditional distribution of $h(\mathbf{X}, S)|S = s$. Most of fair representation learning algorithms consist of

two steps. As a first step, we select a suitable deviance measure for the dissimilarity of $\mathcal{P}_0^h$ and $\mathcal{P}_1^h$. Then, the second step is to find a representation that minimizes the deviance measure. For the deviance measure, Madras et al. (2018) uses the KL divergence, Deka and Sutherland (2023) uses the Maximum Mean Discrepancy (MMD), and Kim et al. (2022a) uses the SIPM.

5.2. Learning a fair representation by use of the ReLU-IPM

Let $\mathcal{H}, \mathcal{G}$ and ℓ be a given set of representations, a given set of prediction heads on $\mathcal{Z}$ and a given loss function on the output space, respectively. We consider the algorithm that learns a fair representation h by minimizing $\mathbb{E}\ell(g \circ h(\mathbf{X}, S), Y)$ subject to $d_{\mathcal{F}}(\mathcal{P}_0^h, \mathcal{P}_1^h) \leq \delta$ with respect to $h \in \mathcal{H}$ and $g \in \mathcal{G}$ for a prespecified discriminator class $\mathcal{F}$ of functions from $\mathcal{Z}$ to $\mathbb{R}$. Here, $\delta > 0$ is a given constant. We refer to the case where $\mathcal{F} = \mathcal{F}_{\text{ReLU}}$ as *ReLU-FRL*. For given data, we replace the probability measures and expectations by their empirical counterparts.

A fair representation can be considered as fairly transformed data since it would be expected that $g \circ h$ is fair for a reasonable prediction head g . We say that a function f is groupwisely η -fair with respect to a fair function $\phi: \mathbb{R} \rightarrow \mathbb{R}$ (Madras et al., 2018; Chuang and Mroueh, 2021; Kim et al., 2022a) if $\Delta\text{DP}_{\phi}(f) \leq \eta$ where

$$\Delta\text{DP}_{\phi}(f) = |\mathbb{E}(\phi \circ f(\mathbf{X}, S)|S = 0) - \mathbb{E}(\phi \circ f(\mathbf{X}, S)|S = 1)|.$$

The existing well-known measures for group fairness can be expressed by choosing a corresponding fair function ϕ . For example, the conventional measure ΔDP (Calders et al., 2009; Feldman et al., 2015; Donini et al., 2018; Agarwal et al., 2019; Zafar et al., 2019) is ΔDP_{ϕ} with $\phi(t) = \mathbb{I}(t \geq 1/2)$. Another popularly used measure $\Delta\bar{\text{DP}}$, which is the disparity between prediction scores (Madras et al., 2018; Chuang and Mroueh, 2021; Kim et al., 2022a), is defined with $\phi(t) = t$. ΔSDP , which is a distributional gap between prediction scores, is defined as $\Delta\text{SDP} = \mathbb{E}_{\tau \in \text{Unif}(0,1)} \Delta\text{DP}_{\phi_{\tau}}$, where $\phi_{\tau}(t) = \mathbb{I}(t \geq \tau)$ (Chzhen et al., 2020; Jiang et al., 2020; Silvia et al., 2020; Barata et al., 2021). Moreover, ΔDP_{ϕ} has been used as a surrogate for ΔDP , with $\phi(t) = (1 - t)_{+}$ and $\phi(t) = \text{sig}(t)$ (Wu et al., 2019; Kim et al., 2022b; Kim and Jeong, 2022; Yao et al., 2023).

A given fair representation h is said to be (groupwisely) η -fair on a given set $\mathcal{G}$ of prediction heads if $g \circ h$ is η -fair for all $g \in \mathcal{G}$. If the IPM with a discriminator class $\mathcal{F}$ is used for the deviance measure of FRL, it is easy to

see that the learned fair representation h is η -fair as long as $\mathcal{G} \subset \mathcal{F}$. However, for the ReLU-FRL, the learned fair representation h can be η -fair on a much larger class of prediction heads than $\mathcal{F}_{\text{ReLU}}$ thanks to Theorem 2, whose results are rigorously stated in the following theorem.

Theorem 5 (Level of group fairness). *For any Lipschitz function ϕ , there exists a constant $c > 0$ such that*

(i) if $\beta < \frac{d+3}{2}$, for every $g \in \mathcal{H}_d^\beta$, we have

$$\Delta\text{DP}_\phi(g \circ h) \leq c \cdot d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}_0^h, \mathcal{P}_1^h)^{\frac{2\beta}{d+3}},$$

(ii) and if $\beta \geq \frac{d+3}{2}$, for every $g \in \mathcal{H}_d^\beta$, we have

$$\Delta\text{DP}_\phi(g \circ h) \leq c \cdot d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}_0^h, \mathcal{P}_1^h).$$

5.3. Numerical Experiments

5.3.1. Settings

Datasets. We analyze three real benchmark tabular datasets: ADULT, DUTCH, and CRIME, which are known to have unfair bias with respect to specific sensitive attributes. Table D.3 in Appendix D.1 provides a summary of key characteristics for these three datasets. We randomly split each dataset five times into training and test sets with a ratio of 8:2, and report the average performances on the five test datasets.

Model architectures. For the representation encoder h , we use a two-layered MLP with the ReLU activation, where the number of nodes in each hidden layer is set to be equal to the input dimensions. For the prediction head g , we consider a linear logistic regression model and a single-layer neural network with the sigmoid activation function and 100 nodes.

Baseline algorithms. For baselines, we consider the following four FRL algorithms: (i) SIPM-LFR (Kim et al., 2022a) and (ii) MMD-B-Fair (Deka and Sutherland, 2023) which use the SIPM and MMD to measure the deviance of the distributions of the representation corresponding to the two sensitive groups, respectively, (iii) LAFTR (Madras et al., 2018) which employs an adversarial learning on the representation space, and (iv) Hölder-FRL which uses the Hölder-IPM whose discriminators are approximated by DNNs with one hidden layer. In Appendix D.2, we provide implementation details.

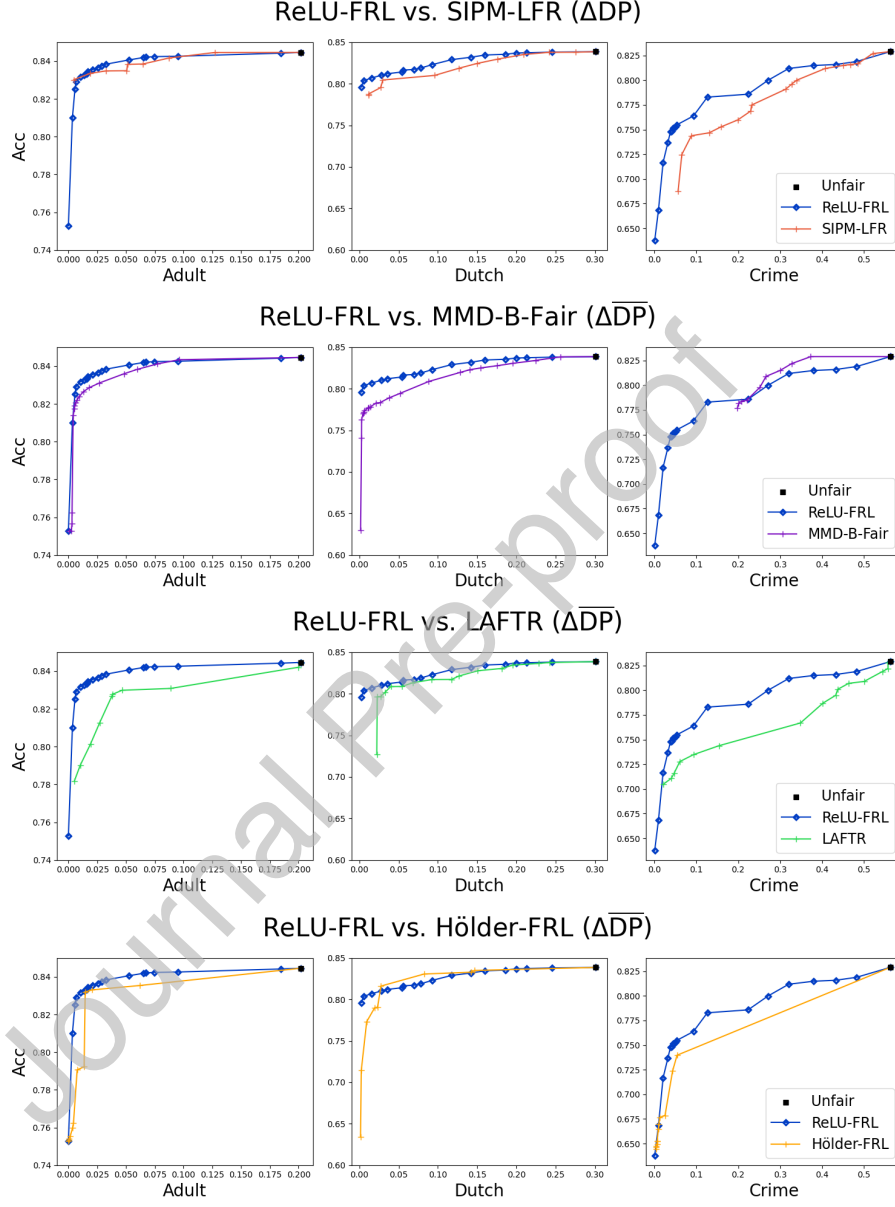


Figure 1: **Linear logistic prediction head:** Pareto-front lines of fairness level $\Delta\overline{DP}$ and ACC. (Left) ADULT, (Center) DUTCH, (Right) CRIME.

5.3.2. Results

For the comparison of empirical performances, we consider the trade-off between accuracy (ACC) and fairness levels ($\Delta\overline{DP}$, ΔDP , ΔSDP), following the approach used in previous works (Madras et al., 2018; Kim et al., 2022a; Deka and Sutherland, 2023). Figure 1 compares the ReLU-FRL with the other baselines through the Pareto-front trade-off graphs between fairness level $\Delta\overline{DP}$ and ACC across the three datasets (ADULT, DUTCH, CRIME), with logistic prediction heads. The Pareto-front lines for the other fairness measures including ΔDP , ΔSDP vs ACC along with different prediction head g are presented in Appendix D.3. The results show that the ReLU-FRL is comparable to the other baselines. In particular, The ReLU-FRL outperforms the other baselines for ADULT, even though the margins are not large. In addition, it is observed that the MMD-B-fair has a difficulty in achieving the fairness level below a certain level.

5.4. Application to fair image classification

In this subsection, we analyze image data to illustrate that the ReLU-FRL works well not only for tabular data but also for complex AI problems.

Datasets. We analyze a benchmark image dataset: CELEBA, where facial attributes are annotated with binary labels. We consider ‘Gender (Male)’ as the sensitive attribute and ‘Smile’ as the target variable. Table D.3 in Appendix D.1 provides a summary of key characteristics of CELEBA. A pretrained ResNet18 is used to extract 512-dimensional feature embeddings. We randomly split the dataset five times into training and test datasets with a ratio of 8:2, and report the average performance over the five test datasets.

Model architectures. We use the same model architectures as in the tabular experiments on the top of the pretrained ResNet18. That is, for the representation encoder h , we use a two-layered MLP with the ReLU activation, where the number of nodes in each hidden layer is set to be equal to the input dimensions, and the prediction head g is defined by a linear logistic regression model.

Baseline algorithms. We use the same baselines, training procedure and hyperparameter settings as in the tabular experiments unless otherwise specified. For implementation details, see Appendix D.2.

Result. Figure 2 compares the ReLU-FRL with other baselines through Pareto-front trade-off lines between the fairness levels ($\Delta\overline{DP}$, ΔDP , ΔSDP) and ACC on CELEBA with logistic prediction heads. In particular, the ReLU-FRL is compared favorably to the other baselines across all fairness levels, while the MMD-B-Fair and LAFTR consistently underperform compared to the other baselines.

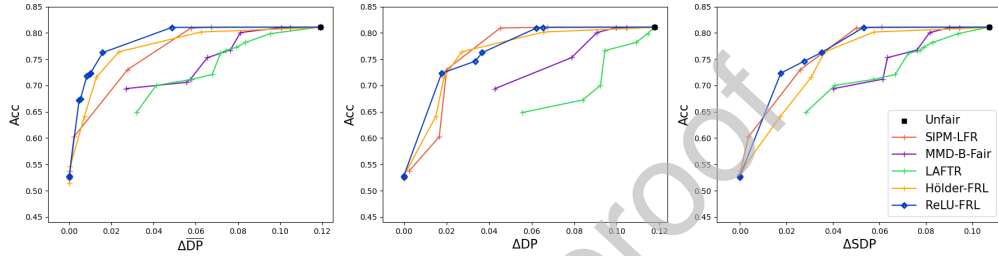


Figure 2: Pareto-front lines on CELEBA of fairness level ($\Delta\overline{DP}$, ΔDP , ΔSDP) and ACC.

6. Conclusion

In this study, we proposed a new parametric IPM, called the ReLU-IPM, and established its desirable theoretical properties. Numerical experiments further demonstrated its usefulness for causal effect estimation and learning fair neural networks. Notably, compared with the Hölder-IPM, the ReLU-IPM has several advantages: (i) it has similar theoretical guarantees, in particular, when the discriminator class is sufficiently smooth, (ii) it is simpler to implement, (iii) it requires fewer hyperparameters, and (iv) it demonstrates comparable or superior performance in numerical experiments. Compared with the SIPM, the ReLU-IPM provides stronger guarantees, including a parametric convergence rate for ATT estimation and group-fairness guarantees for a larger class of prediction heads.

Several directions remain for future research. First, although we established a parametric convergence rate for the ATT estimator, proving its conjectured semiparametric efficiency remains an important direction for future research. Second, other activation functions could be considered in place of ReLU. In particular, leaky ReLU would be a promising alternative because its additional parameter could be selected in a data-adaptive manner to provide additional flexibility. Third, the ReLU-IPM may provide a way to explain how two given distributions differ. For example, by examining

the distributions of $\hat{\theta}^\top \mathbf{X}$ and $\hat{\theta}^\top \mathbf{Y}$, where $\hat{\theta}$ is the parameter at which the ReLU-IPM is evaluated, one may gain insight into the difference between $\mathcal{P}$ and $\mathcal{Q}$.

The ReLU-IPM may further serve as a simple discrepancy measure in counterfactual learning with deep neural networks. One possible application is counterfactual image generation (Jerzak et al., 2023), where distributions of generated and target counterfactual samples need to be compared or regularized. This would provide a link between the proposed theory and neural-network-based causal learning and generative modeling.

Finally, ReLU-IPM-based fair representation learning could be applied to robot control and physical AI systems, where fairness has recently received increasing attention (Jabbari et al., 2017; Mehrabi et al., 2021; Londoño et al., 2024; Fosch-Villaronga et al., 2025; Kattinig et al., 2024). A fair representation head could be added to the top of a pretrained model to improve the fairness of downstream decisions. In turn, a high-quality fair representation head could be learned by our proposed FRL algorithm with less energy and data (since the discriminator class of the ReLU-IPM is parametric).

Appendix A. Definitions and Auxiliary lemmas

In this section, we provide the mathematical tools used in this paper.

Definition 1 (Bounded differences property). *A function $f : \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n \rightarrow \mathbb{R}$ is said to satisfy the bounded differences property if substituting the value of the i -th coordinate $\mathbf{x}_i$ changes the value of f by at most c_i . More formally, if there exist constants $c_1, c_2, \dots, c_n$ such that for all $i \in \{1, \dots, n\}$, and all $\mathbf{x}_1 \in \mathcal{X}_1, \mathbf{x}_2 \in \mathcal{X}_2, \dots, \mathbf{x}_n \in \mathcal{X}_n$,*

$$\sup_{\mathbf{x}'_i \in \mathcal{X}'_i} \left| f(\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathbf{x}_i, \mathbf{x}_{i+1}, \dots, \mathbf{x}_n) - f(\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathbf{x}'_i, \mathbf{x}_{i+1}, \dots, \mathbf{x}_n) \right| \leq c_i.$$

Definition 2 (Rademacher random variable). *A random variable $X \in \{-1, 1\}$ is called a Rademacher random variable if $\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = \frac{1}{2}$.*

Definition 3 ($L_p(\mathcal{Q})$ -norm). *Let $\mathcal{Q}$ be a measure on a measurable space $(\mathcal{X}, \mathcal{A})$ and $L_p(\mathcal{Q}) = \{f : \mathcal{X} \rightarrow \mathbb{R} : \int |f|^p d\mathcal{Q} < \infty\}$. For $f \in L_p(\mathcal{Q})$, the $L_p(\mathcal{Q})$ -norm is defined as $\|f\|_{p, \mathcal{Q}} := (\int |f|^p d\mathcal{Q})^{1/p}$. When $p = 2$, we omit the subscript 2, i.e., $\|\cdot\|_{\mathcal{Q}} = \|\cdot\|_{2, \mathcal{Q}}$.*

Definition 4 ($L_p(\mathcal{Q}_n)$ -norm). *Let $\mathcal{Q}_n$ be the empirical measure of $z_1, \dots, z_n$. We define the $L_p(\mathcal{Q}_n)$ -norm by $\|f\|_{p, \mathcal{Q}_n} := (\frac{1}{n} \sum_{i=1}^n |f(z_i)|^p)^{1/p}$. For $p = 2$, we omit the subscript 2, writing $\|\cdot\|_{\mathcal{Q}_n}$ instead of $\|\cdot\|_{2, \mathcal{Q}_n}$.*

Definition 5 (ϵ -covering number and ϵ -entropy). *Let $N_p(\mathcal{F}, \|\cdot\|_{\mathcal{Q}}, \epsilon)$ be the ϵ -covering number, defined as the smallest N such that for every $\epsilon > 0$ and a class of functions $\mathcal{F} \subset L_2(\mathcal{Q})$, there exist functions $f_1, \dots, f_N \in L_2(\mathcal{Q})$ such that $\mathcal{F}$ can be covered by N balls of radius ϵ centered at $f_1, \dots, f_N$ in $L_2(\mathcal{Q})$. For $\epsilon > 0$ and a class of functions $\mathcal{F} \subset L_2(\mathcal{Q})$, let $H(\mathcal{F}, \|\cdot\|_{\mathcal{Q}}, \epsilon) = \log N_p(\mathcal{F}, \|\cdot\|_{\mathcal{Q}}, \epsilon)$ be the ϵ -entropy of $\mathcal{F}$ with respect to the $L_2(\mathcal{Q})$ -metric.*

Lemma 1 (McDiarmid's inequality (McDiarmid, 1989)). *Let $f : \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n \rightarrow \mathbb{R}$ satisfy the bounded differences property with bounds $c_1, c_2, \dots, c_n$, and let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be independent random vectors where $\mathbf{X}_i \in \mathcal{X}_i$ for all i . Then, for any $\epsilon > 0$,*

$$\begin{aligned} \mathbb{P}\{f(\mathbf{X}_1, \dots, \mathbf{X}_n) - \mathbb{E}[f(\mathbf{X}_1, \dots, \mathbf{X}_n)] \geq \epsilon\} &\leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}\right), \\ \mathbb{P}\{f(\mathbf{X}_1, \dots, \mathbf{X}_n) - \mathbb{E}[f(\mathbf{X}_1, \dots, \mathbf{X}_n)] \leq -\epsilon\} &\leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}\right), \end{aligned}$$

and as an immediate consequence,

$$\mathbb{P}\{|f(\mathbf{X}_1, \dots, \mathbf{X}_n) - \mathbb{E}[f(\mathbf{X}_1, \dots, \mathbf{X}_n)]| \geq \epsilon\} \leq 2 \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}\right).$$

Lemma 2 (Ledoux-Talagrand contraction inequality (Ledoux and Talagrand, 1991)). *For a compact set $\mathcal{T}$, let $\mathbf{X}_1, \dots, \mathbf{X}_m$ be i.i.d random vectors whose real-valued components are indexed by $\mathcal{T}$, i.e., $\mathbf{X}_i = (\mathbf{X}_{i,s})_{s \in \mathcal{T}}$. Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be a 1-Lipschitz function such that $\phi(0) = 0$. Let $\epsilon_1, \dots, \epsilon_m$ be independent Rademacher random variables. Then*

$$\mathbb{E} \left[\sup_{s \in \mathcal{T}} \left| \sum_{i=1}^m \epsilon_i \phi(\mathbf{X}_{i,s}) \right| \right] \leq 2 \mathbb{E} \left[\sup_{s \in \mathcal{T}} \left| \sum_{i=1}^m \epsilon_i \mathbf{X}_{i,s} \right| \right].$$

Lemma 3 (Lemma 8.4 of Van de Geer (2000)). *Let $W_1, \dots, W_n$ be independent random variables with mean zero and let $z_1, \dots, z_n$ be a set of points. We denote the empirical measure of $z_1, \dots, z_n$ by Q_n . Suppose that for some $\alpha \in (0, 2)$, $A > 0$, R , K , and σ_0 , there exists some function class $\mathcal{F}$ such that (i) $H(\mathcal{F}, \|\cdot\|_{Q_n}, \epsilon) \leq A\epsilon^{-\alpha}$ holds for all $\epsilon > 0$, (ii) $\sup_{f \in \mathcal{F}} \|f\|_{Q_n} \leq R$, (iii) $\max_{i=1, \dots, n} K^2 \left(\mathbb{E} e^{|W_i|^2/K^2} - 1 \right) \leq \sigma_0^2$. Then there exists a constant c depending on A, α, R, K , and σ_0 such that we have*

$$\mathbb{P} \left(\sup_{f \in \mathcal{F}} \frac{\left| \frac{1}{\sqrt{n}} \sum_{i=1}^n W_i f(z_i) \right|}{\|f\|_{Q_n}^{1-\alpha/2}} \geq T \right) \leq c \exp \left(-\frac{T^2}{c^2} \right),$$

for all $T \geq c$.

Appendix B. Proofs of Theoretical Results

Appendix B.1. Proof of Proposition 1

We begin with the following lemma, which plays a key role in the proof of Proposition 1.

Lemma 4. *For two random vectors $\mathbf{X} \sim \mathcal{P}$ and $\mathbf{Y} \sim \mathcal{Q}$, if $d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}) = 0$, then*

$$\sup_{\mathbf{c} \in \mathbb{S}^{d-1}} \sup_{t \in \mathbb{R}} |\mathbb{P}(\mathbf{c}^\top \mathbf{X} \leq t) - \mathbb{P}(\mathbf{c}^\top \mathbf{Y} \leq t)| = 0. \quad (\text{B.1})$$

Proof of Lemma 4. By the definition of the ReLU-IPM, $d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}) = 0$ implies

$$|\mathbb{E}(\mathbf{c}^\top \mathbf{X})_+ - \mathbb{E}(\mathbf{c}^\top \mathbf{Y})_+| = 0, \quad (\text{B.2})$$

for any $\mathbf{c} \in \mathbb{S}^{d-1}$. For given $\mathbf{c} \in \mathbb{S}^{d-1}$, we consider $t \in \mathbb{R}$ such that random variables $\mathbf{c}^\top \mathbf{X}$ and $\mathbf{c}^\top \mathbf{Y}$ do not have a point mass at t . For $\delta > 0$, let $\Delta_\delta(\mathbf{X}) := (\mathbf{c}^\top \mathbf{X} - t + \delta)_+ - (\mathbf{c}^\top \mathbf{X} - t)_+$. Then, we can write

$$\Delta_\delta(\mathbf{X}) = (\mathbf{c}^\top \mathbf{X} - t + \delta) \cdot \mathbb{I}(t - \delta < \mathbf{c}^\top \mathbf{X} \leq t) + \delta \cdot \mathbb{I}(t < \mathbf{c}^\top \mathbf{X}). \quad (\text{B.3})$$

From (B.3), we obtain

$$\begin{aligned} & \left| \mathbb{P}(\mathbf{c}^\top \mathbf{X} > t) - \frac{1}{\delta} \mathbb{E} \Delta_\delta(\mathbf{X}) \right| \\ &= \left| \mathbb{P}(\mathbf{c}^\top \mathbf{X} > t) - \mathbb{E} \left(\frac{\mathbf{c}^\top \mathbf{X} - t + \delta}{\delta} \cdot \mathbb{I}(t - \delta < \mathbf{c}^\top \mathbf{X} \leq t) \right) - \mathbb{P}(\mathbf{c}^\top \mathbf{X} > t) \right|, \\ &\leq \mathbb{P}(t - \delta < \mathbf{c}^\top \mathbf{X} \leq t), \end{aligned} \quad (\text{B.4})$$

where we use $0 \leq \mathbf{c}^\top \mathbf{X} - t + \delta \leq \delta$ for the last inequality. Similarly, for $\Delta_\delta(\mathbf{Y}) := (\mathbf{c}^\top \mathbf{Y} - t + \delta)_+ - (\mathbf{c}^\top \mathbf{Y} - t)_+$,

$$\left| \mathbb{P}(\mathbf{c}^\top \mathbf{Y} > t) - \frac{1}{\delta} \mathbb{E} \Delta_\delta(\mathbf{Y}) \right| \leq \mathbb{P}(t - \delta < \mathbf{c}^\top \mathbf{Y} \leq t). \quad (\text{B.5})$$

Therefore, from (B.2), (B.4) and (B.5), we get

$$\begin{aligned}
& \left| \mathbb{P}(\mathbf{c}^\top \mathbf{X} \leq t) - \mathbb{P}(\mathbf{c}^\top \mathbf{Y} \leq t) \right| \\
&= \left| \mathbb{P}(\mathbf{c}^\top \mathbf{X} > t) - \mathbb{P}(\mathbf{c}^\top \mathbf{Y} > t) \right| \\
&\leq \left| \frac{1}{\delta} \mathbb{E} \Delta_\delta(\mathbf{X}) - \frac{1}{\delta} \mathbb{E} \Delta_\delta(\mathbf{Y}) \right| + \mathbb{P}(t - \delta < \mathbf{c}^\top \mathbf{X} \leq t) + \mathbb{P}(t - \delta < \mathbf{c}^\top \mathbf{Y} \leq t) \\
&\leq \frac{1}{\delta} \left| \mathbb{E}(\mathbf{c}^\top \mathbf{X} - t + \delta)_+ - \mathbb{E}(\mathbf{c}^\top \mathbf{X} - t)_+ - \mathbb{E}(\mathbf{c}^\top \mathbf{Y} - t + \delta)_+ + \mathbb{E}(\mathbf{c}^\top \mathbf{Y} - t)_+ \right| \\
&\quad + \mathbb{P}(t - \delta < \mathbf{c}^\top \mathbf{X} \leq t) + \mathbb{P}(t - \delta < \mathbf{c}^\top \mathbf{Y} \leq t) \\
&= \mathbb{P}(t - \delta < \mathbf{c}^\top \mathbf{X} \leq t) + \mathbb{P}(t - \delta < \mathbf{c}^\top \mathbf{Y} \leq t).
\end{aligned}$$

Since $\delta > 0$ is arbitrary, we obtain

$$\begin{aligned}
& \left| \mathbb{P}(\mathbf{c}^\top \mathbf{X} \leq t) - \mathbb{P}(\mathbf{c}^\top \mathbf{Y} \leq t) \right| \\
&\leq \lim_{\delta \downarrow 0} \mathbb{P}(t - \delta < \mathbf{c}^\top \mathbf{X} \leq t) + \lim_{\delta \downarrow 0} \mathbb{P}(t - \delta < \mathbf{c}^\top \mathbf{Y} \leq t) \\
&= 0.
\end{aligned}$$

For the case where either $\mathbf{c}^\top \mathbf{X}$ or $\mathbf{c}^\top \mathbf{Y}$ has a point mass at t , we can construct a sequence $\{t_j\}_{j=1}^\infty$ such that $t_j \downarrow t$ and neither $\mathbf{c}^\top \mathbf{X}$ nor $\mathbf{c}^\top \mathbf{Y}$ has a point mass at $\{t_j\}_{j=1}^\infty$. Since $\mathbb{P}(\mathbf{c}^\top \mathbf{X} \leq \cdot)$ and $\mathbb{P}(\mathbf{c}^\top \mathbf{Y} \leq \cdot)$ is right-continuous,

$$\left| \mathbb{P}(\mathbf{c}^\top \mathbf{X} \leq t) - \mathbb{P}(\mathbf{c}^\top \mathbf{Y} \leq t) \right| = \lim_{j \rightarrow \infty} \left| \mathbb{P}(\mathbf{c}^\top \mathbf{X} \leq t_j) - \mathbb{P}(\mathbf{c}^\top \mathbf{Y} \leq t_j) \right| = 0,$$

and the proof is done. $\square$

Proof of Proposition 1. ($\Leftarrow$) It is obvious since for any $f \in \mathcal{F}_{\text{ReLU}}$, we have

$$\int f(\mathbf{z})(d\mathcal{P}(\mathbf{z}) - d\mathcal{Q}(\mathbf{z})) = 0.$$

($\Rightarrow$) Lemma 4 implies $\mathbf{c}^\top \mathbf{X} \stackrel{d}{=} \mathbf{c}^\top \mathbf{Y}$ for all $\mathbf{c} \in \mathbb{R}^d$, which, in turn, implies $\mathcal{P} \equiv \mathcal{Q}$ by the uniqueness of the characteristic function. $\square$

Appendix B.2. Proof of Theorem 2

We use the following corollary to prove Theorem 2.

Lemma 5 (Corollary 2.4 of Yang and Zhou (2025)). *Let $d \geq 3$ and $\beta > 0$. For any $h \in \mathcal{H}_d^\beta$,*

- (i) *if $\beta \geq \frac{d+3}{2}$, then for any sufficiently large L and $M \asymp \sqrt{\log L}$, there exist $f_1, f_2, \dots, f_L \in \mathcal{F}_{\text{ReLU}}$ and $a_1, a_2, \dots, a_L$ such that $\sum_{i=1}^L |a_i| \leq M$ and*

$$\left\| h - \sum_{i=1}^L a_i f_i \right\|_\infty \lesssim L^{-\frac{d+3}{2d}} \sqrt{\log L},$$

where M is a constant depending on β and d ,

- (ii) *if $\beta < \frac{d+3}{2}$, then for any sufficiently large L and M , there exist $f_1, f_2, \dots, f_L \in \mathcal{F}_{\text{ReLU}}$ and $a_1, a_2, \dots, a_L$ such that $\sum_{i=1}^L |a_i| \leq M$ and*

$$\left\| h - \sum_{i=1}^L a_i f_i \right\|_\infty \lesssim L^{-\frac{\beta}{d}} \vee M^{-\frac{2\beta}{d+3-2\beta}},$$

Thus, the upper bound becomes $\mathcal{O}(L^{-\frac{\beta}{d}})$ holds when $M \gtrsim L^{-\frac{d+3-2\beta}{2d}}$.

Remark 3. *In case (i), if $\beta > \frac{d+3}{2}$ or $\beta = \frac{d+3}{2}$ is an even integer, then for any sufficiently large L , the $\sqrt{\log L}$ term vanishes. Consequently, the approximation bound becomes $L^{-\frac{d+3}{2d}}$.*

Proof of Theorem 2. (i) $\beta \geq \frac{d+3}{2}$: By Lemma 5, if $d > 3$, for all $h \in \mathcal{H}_d^\beta$, when $\beta \geq \frac{d+3}{2}$, there exists a constant M depending on d and β such that

$$\left\| h - \sum_{i=1}^L a_i f_i \right\|_\infty \lesssim L^{-\frac{d+3}{2d}} \sqrt{\log L},$$

and $\sum_{i=1}^L |a_i| \leq M$. Then we have

$$\begin{aligned} & \left| \int h(\mathbf{X}) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| \\ &= \left| \int \left(h(\mathbf{X}) - \sum_{i=1}^L a_i f_i(\mathbf{X}) + \sum_{i=1}^L a_i f_i(\mathbf{X}) \right) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| \end{aligned} \tag{B.6}$$

$$\begin{aligned}
&\leq \left| \int \sum_{i=1}^L a_i f_i(\mathbf{X}) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| \\
&\quad + \left| \int \left(h(\mathbf{X}) - \sum_{i=1}^L a_i f_i(\mathbf{X}) \right) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| \\
&\lesssim \left| \int \sum_{i=1}^L a_i f_i(\mathbf{X}) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| + L^{-\frac{d+3}{2d}} \sqrt{\log L} \\
&\leq \sum_{i=1}^L |a_i| \left| \int f_i(\mathbf{X}) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| + L^{-\frac{d+3}{2d}} \sqrt{\log L} \\
&\leq M \cdot d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}) + L^{-\frac{d+3}{2d}} \sqrt{\log L} \\
&\lesssim d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}),
\end{aligned}$$

where the last inequality holds with sufficiently large L .

(ii) $\beta < \frac{d+3}{2}$: Let $d \in \mathbb{N}$ and $\beta > 0$. By Lemma 5, for all $h \in \mathcal{H}_d^\beta$, there exist $a_1, \dots, a_L \in \mathbb{R}$ and $f_1, \dots, f_L \in \mathcal{F}_{\text{ReLU}}$ such that $\|h - \sum_{i=1}^L a_i f_i\|_\infty \lesssim L^{-\frac{\beta}{d}} \vee M^{-\frac{2\beta}{d-2\beta+3}}$ and $\sum_{i=1}^L |a_i| \leq M$. Then we have

$$\begin{aligned}
&\left| \int h(\mathbf{X}) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| \\
&= \left| \int \left(h(\mathbf{X}) - \sum_{i=1}^L a_i f_i(\mathbf{X}) + \sum_{i=1}^L a_i f_i(\mathbf{X}) \right) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| \\
&\leq \left| \int \sum_{i=1}^L a_i f_i(\mathbf{X}) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| \\
&\quad + \left| \int \left(h(\mathbf{X}) - \sum_{i=1}^L a_i f_i(\mathbf{X}) \right) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| \\
&\lesssim \left| \int \sum_{i=1}^L a_i f_i(\mathbf{X}) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| + \left(L^{-\frac{\beta}{d}} \vee M^{-\frac{2\beta}{d-2\beta+3}} \right)
\end{aligned} \tag{B.7}$$

$$\begin{aligned}
&\leq \sum_{i=1}^L |a_i| \left| \int f_i(\mathbf{X}) (d\mathcal{P}(\mathbf{X}) - d\mathcal{Q}(\mathbf{X})) \right| + \left(L^{-\frac{\beta}{d}} \vee M^{-\frac{2\beta}{d-2\beta+3}} \right) \\
&\leq M \cdot d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}) + \left(L^{-\frac{\beta}{d}} \vee M^{-\frac{2\beta}{d-2\beta+3}} \right).
\end{aligned}$$

Thus, we can complete the proof if we choose $M = d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q})^{-\frac{d-2\beta+3}{d+3}}$ for sufficiently large L . $\square$

Appendix B.3. Proof of Theorem 3

Proof. We consider the case of $n \neq m$. To prove Theorem 3, we will show

$$\begin{aligned}
&\mathbb{P} \left(\left| d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}) - d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_n, \hat{\mathcal{Q}}_m) \right| - 2 \left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}} \right) > \epsilon \right) \\
&\leq 2 \exp \left(-\frac{\epsilon^2 nm}{8(n+m)} \right). \quad (\text{B.8})
\end{aligned}$$

Let us begin with the absolute term on the left-hand side of (B.8).

$$\begin{aligned}
&\left| d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}, \mathcal{Q}) - d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_n, \hat{\mathcal{Q}}_m) \right| \\
&= \left| \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \mathbb{E}_{\mathbf{X} \sim \mathcal{P}} f - \mathbb{E}_{\mathbf{Y} \sim \mathcal{Q}} f \right| - \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{X}_i) - \frac{1}{m} \sum_{i=1}^m f(\mathbf{Y}_i) \right| \right| \\
&\leq \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \mathbb{E}_{\mathbf{X} \sim \mathcal{P}} f - \mathbb{E}_{\mathbf{Y} \sim \mathcal{Q}} f - \frac{1}{n} \sum_{i=1}^n f(\mathbf{X}_i) + \frac{1}{m} \sum_{i=1}^m f(\mathbf{Y}_i) \right| \\
&:= \Phi(\mathcal{P}, \mathcal{Q}, \mathbf{X}^n, \mathbf{Y}^m), \quad (\text{B.9})
\end{aligned}$$

where $\mathbf{X}^n = (\mathbf{X}_1, \dots, \mathbf{X}_n)$ and $\mathbf{Y}^m$ is defined similarly. Define $\mathbf{X}'_{(i)}$ as a copy of $\mathbf{X}^n$ differing only in its i -th element, for $i = 1, \dots, n$, and let $\mathbf{Y}'_{(j)}$ be defined analogously for $j = 1, \dots, m$. Then we have

$$\begin{aligned}
& \Phi(\mathcal{P}, \mathcal{Q}, \mathbf{X}^n, \mathbf{Y}^m) - \Phi(\mathcal{P}, \mathcal{Q}, \mathbf{X}'_{(i)}, \mathbf{Y}^m) \\
& \leq \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \mathbb{E}_{\mathbf{X} \sim \mathcal{P}} f - \mathbb{E}_{\mathbf{Y} \sim \mathcal{Q}} f - \frac{1}{n} \sum_{i=1}^n f(\mathbf{X}_i) + \frac{1}{m} \sum_{i=1}^m f(\mathbf{Y}_i) \right| \\
& \quad - \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \mathbb{E}_{\mathbf{X} \sim \mathcal{P}} f - \mathbb{E}_{\mathbf{Y} \sim \mathcal{Q}} f - \frac{1}{n} \left(\sum_{k \neq i} f(\mathbf{X}_k) + f(\mathbf{X}'_i) \right) + \frac{1}{m} \sum_{i=1}^m f(\mathbf{Y}_i) \right| \\
& \leq \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| -\frac{1}{n} \sum_{i=1}^n f(\mathbf{X}_i) + \frac{1}{n} \left(\sum_{k \neq i} f(\mathbf{X}_k) + f(\mathbf{X}'_i) \right) \right| \\
& = \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \frac{1}{n} f(\mathbf{X}'_i) - \frac{1}{n} f(\mathbf{X}_i) \right| \\
& \leq \frac{2}{n} \sup_{f \in \mathcal{F}_{\text{ReLU}}} \|f\|_{\infty} \\
& \leq \frac{4}{n},
\end{aligned}$$

where the last inequality follows from $\|f(\mathbf{z})\|_2 \leq \|\mathbf{z}^\top \boldsymbol{\theta} + \mu\|_2 = |\mathbf{z}^\top \boldsymbol{\theta} + \mu| \leq 2$, for all $\mathbf{z} \in \mathbb{B}^d$, $\boldsymbol{\theta} \in \mathbb{S}^{d-1}$ and $\mu \in [-1, 1]$. Similarly, $\Phi(\mathcal{P}, \mathcal{Q}, \mathbf{X}^n, \mathbf{Y}^m) - \Phi(\mathcal{P}, \mathcal{Q}, \mathbf{X}^n, \mathbf{Y}'_{(j)}) \leq 4/m$. Applying Lemma 1 (McDiarmid's inequality) with $c_i = 4/n$ ($i = 1, \dots, n$) for $\mathbf{X}$ and $c_j = 4/m$ ($j = 1, \dots, m$) for $\mathbf{Y}$ (that is, a denominator in the exponent of Lemma 1 of $16(n+m)/(nm)$), we obtain

$$\mathbb{P}(\Phi(\mathcal{P}, \mathcal{Q}, \mathbf{X}^n, \mathbf{Y}^m) - \mathbb{E}_{\mathbf{X}, \mathbf{Y}} \Phi(\mathcal{P}, \mathcal{Q}, \mathbf{X}^n, \mathbf{Y}^m) > \epsilon) \leq 2 \exp \left(-\frac{\epsilon^2 nm}{8(n+m)} \right). \quad (\text{B.10})$$

The proof would be done if we derive the upper bound of the expectation of $\Phi(\mathcal{P}, \mathcal{Q}, \mathbf{X}^n, \mathbf{Y}^m)$ as follows:

$$\mathbb{E}_{\mathbf{X}, \mathbf{Y}} \Phi(\mathcal{P}, \mathcal{Q}, \mathbf{X}^n, \mathbf{Y}^m) \leq 4 \left(1 + \sqrt{\frac{2}{\pi}} \right) \left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}} \right).$$

Let $\varepsilon_1, \dots, \varepsilon_n$ be independent Radamacher random variables, i.e., $\mathbb{P}(\varepsilon_i = -1) = \mathbb{P}(\varepsilon_i = 1) = 1/2$. The empirical Radamacher average (Rademacher complexity) of a function class $\mathcal{F}$ is defined as

$$\widehat{\mathcal{R}}_n(\mathcal{F}) := \mathbb{E}_{\varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(\mathbf{X}_i) \right| \middle| \mathbf{X}_1, \dots, \mathbf{X}_n \right],$$

where $\mathbb{E}_\varepsilon$ denotes the expectation over all ε_i . Since $\{\sum_{i=1}^n \varepsilon_i\}$ is a simple symmetric random walk, its expected absolute distance after n steps is given by (Grünbaum, 1960; Mosteller et al., 1961; Hughes, 1995; König et al., 1999)

$$\mathbb{E} \left| \sum_{i=1}^n \varepsilon_i \right| \leq \sqrt{\frac{2n}{\pi}}. \quad (\text{B.11})$$

Thus, by the Talagrand's Contraction Lemma (Ledoux and Talagrand, 1991) (see Lemma 2 of Appendix A), we have an upper bound of the empirical Rademacher average for the discriminator class $\mathcal{F}_{\text{ReLU}}$ as

$$\begin{aligned} \widehat{\mathcal{R}}_n(\mathcal{F}_{\text{ReLU}}) &= \mathbb{E} \sup_{(\boldsymbol{\theta}, \mu) \in \mathbb{S}^{d-1} \times [-1, 1]} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (\boldsymbol{\theta}^\top \mathbf{X}_i + \mu) \right| \\ &\leq \frac{2}{n} \mathbb{E} \sup_{(\boldsymbol{\theta}, \mu) \in \mathbb{S}^{d-1} \times [-1, 1]} \left| \sum_{i=1}^n \varepsilon_i (\boldsymbol{\theta}^\top \mathbf{X}_i + \mu) \right| \quad (\text{by Lemma 2}) \\ &\leq \frac{2}{n} \mathbb{E} \sup_{(\boldsymbol{\theta}, \mu) \in \mathbb{S}^{d-1} \times [-1, 1]} \left\{ \left| \sum_{i=1}^n \boldsymbol{\theta}^\top (\varepsilon_i \mathbf{X}_i) \right| + \left| \sum_{i=1}^n \varepsilon_i \mu \right| \right\} \\ &\quad (\text{by the triangle inequality}) \\ &\leq \frac{2}{n} \mathbb{E} \sup_{(\boldsymbol{\theta}, \mu) \in \mathbb{S}^{d-1} \times [-1, 1]} \left\{ \|\boldsymbol{\theta}^\top\|_2 \left\| \sum_{i=1}^n \varepsilon_i \mathbf{X}_i \right\|_2 + |\mu| \left| \sum_{i=1}^n \varepsilon_i \right| \right\} \\ &\quad (\text{by Cauchy-Schwarz ineq.}) \\ &\leq \frac{2}{n} \mathbb{E} \left\{ \left\| \sum_{i=1}^n \varepsilon_i \mathbf{X}_i \right\|_2 + \left| \sum_{i=1}^n \varepsilon_i \right| \right\} \\ &\leq \frac{2}{n} \left\{ \mathbb{E} \sqrt{\left\langle \sum_{i=1}^n \varepsilon_i \mathbf{X}_i, \sum_{i=1}^n \varepsilon_i \mathbf{X}_i \right\rangle} + \sqrt{\frac{2n}{\pi}} \right\} \quad (\text{by (B.11)}) \\ &\leq \frac{2}{n} \left\{ \mathbb{E} \sqrt{\sum_{i=1}^n \sum_{j=1}^n \varepsilon_i \varepsilon_j \langle \mathbf{X}_i, \mathbf{X}_j \rangle} + \sqrt{\frac{2n}{\pi}} \right\} \\ &\quad (\text{by the linearity of inner product}) \\ &\leq \frac{2}{n} \left\{ \sqrt{\mathbb{E} \sum_{1 \leq i, j \leq n} \varepsilon_i \varepsilon_j \langle \mathbf{X}_i, \mathbf{X}_j \rangle} + \sqrt{\frac{2n}{\pi}} \right\} \quad (\text{by Jensen's ineq.}) \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{n} \left\{ \sqrt{\sum_{1 \leq i, j \leq n} \mathbb{E}(\varepsilon_i \varepsilon_j) \langle \mathbf{X}_i, \mathbf{X}_j \rangle} + \sqrt{\frac{2n}{\pi}} \right\} \\
&= \frac{2}{n} \left\{ \sqrt{\sum_{i=1}^n \|\mathbf{X}_i\|_2^2} + \sqrt{\frac{2n}{\pi}} \right\} \quad (\text{by properties of } \varepsilon) \\
&\leq \frac{2}{\sqrt{n}} \left(1 + \sqrt{\frac{2}{\pi}} \right),
\end{aligned}$$

where the expectation is taken over $\{\varepsilon_i\}_{i=1}^n$.

Finally, we will derive an upper bound of $\mathbb{E}_{\mathbf{X}, \mathbf{Y}} \Phi(\mathcal{P}, \mathcal{Q}, \mathbf{X}^n, \mathbf{Y}^m)$ in terms of the empirical Radamacher averages by use of the symmetrization technique (van der vaart and Wellner, 2013). Let $\mathbf{X}'^n := (\mathbf{X}'_1, \dots, \mathbf{X}'_n)$ be an independent sample whose distribution is the same as that of $\mathbf{X}^n$, and define $\mathbf{Y}'^m$ similarly. Then we have

$$\begin{aligned}
&\mathbb{E}_{\mathbf{X}, \mathbf{Y}} \Phi(\mathcal{P}, \mathcal{Q}, \mathbf{X}^n, \mathbf{Y}^m) \\
&= \mathbb{E}_{\mathbf{X}^n, \mathbf{Y}^m} \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \mathbb{E}_{\mathbf{X} \sim \mathcal{P}} f(\mathbf{X}) - \mathbb{E}_{\mathbf{Y} \sim \mathcal{Q}} f(\mathbf{Y}) - \frac{1}{n} \sum_{i=1}^n f(\mathbf{X}_i) + \frac{1}{m} \sum_{j=1}^m f(\mathbf{Y}_j) \right| \\
&= \mathbb{E}_{\mathbf{X}^n, \mathbf{Y}^m} \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \mathbb{E}_{\mathbf{X}' \sim \mathcal{P}} \left(\frac{1}{n} \sum_{i=1}^n f(\mathbf{X}'_i) \right) - \frac{1}{n} \sum_{i=1}^n f(\mathbf{X}_i) \right. \\
&\quad \left. - \mathbb{E}_{\mathbf{Y}' \sim \mathcal{Q}} \left(\frac{1}{m} \sum_{j=1}^m f(\mathbf{Y}'_j) \right) + \frac{1}{m} \sum_{j=1}^m f(\mathbf{Y}_j) \right| \\
&\leq \mathbb{E}_{\mathbf{X}^n, \mathbf{Y}^m, \mathbf{X}'^n, \mathbf{Y}'^m} \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{X}'_i) - \frac{1}{n} \sum_{i=1}^n f(\mathbf{X}_i) - \frac{1}{m} \sum_{j=1}^m f(\mathbf{Y}'_j) + \frac{1}{m} \sum_{j=1}^m f(\mathbf{Y}_j) \right| \\
&\quad (\text{by Jensen's ineq.}) \\
&= \mathbb{E}_{\mathbf{X}^n, \mathbf{Y}^m, \mathbf{X}'^n, \mathbf{Y}'^m, \varepsilon, \varepsilon'} \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(\mathbf{X}'_i) - f(\mathbf{X}_i)) + \frac{1}{m} \sum_{j=1}^m \varepsilon'_j (f(\mathbf{Y}'_j) - f(\mathbf{Y}_j)) \right|
\end{aligned} \tag{B.12}$$

$$\begin{aligned}
&\leq \mathbb{E}_{\mathbf{X}^n, \mathbf{X}'^n, \varepsilon} \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(\mathbf{X}'_i) - f(\mathbf{X}_i)) \right| \\
&\quad + \mathbb{E}_{\mathbf{Y}^m, \mathbf{Y}'^m, \varepsilon} \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \frac{1}{m} \sum_{j=1}^m \varepsilon_j (f(\mathbf{Y}'_j) - f(\mathbf{Y}_j)) \right| \quad (\text{by Triangle ineq.}) \\
&= \mathbb{E}_{\mathbf{X}^n, \mathbf{X}'^n} \mathbb{E}_{\varepsilon} \left[\sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(\mathbf{X}'_i) - f(\mathbf{X}_i)) \right| \middle| \mathbf{X}_1, \dots, \mathbf{X}_n \right] \\
&\quad + \mathbb{E}_{\mathbf{Y}^m, \mathbf{Y}'^m} \mathbb{E}_{\varepsilon} \left[\sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \frac{1}{m} \sum_{j=1}^m \varepsilon_j (f(\mathbf{Y}'_j) - f(\mathbf{Y}_j)) \right| \middle| \mathbf{Y}_1, \dots, \mathbf{Y}_m \right] \\
&\leq 2 \left\{ \widehat{\mathcal{R}}_n(\mathcal{F}_{\text{ReLU}}) + \widehat{\mathcal{R}}_m(\mathcal{F}_{\text{ReLU}}) \right\} \\
&\leq 4 \left(1 + \sqrt{2/\pi} \right) (1/\sqrt{n} + 1/\sqrt{m}).
\end{aligned} \tag{B.13}$$

By combining (B.10) with (B.13), we complete the proof of Theorem 3. In addition, the strong consistency of $d_{\mathcal{F}_{\text{ReLU}}}(\widehat{\mathcal{P}}_n, \widehat{\mathcal{Q}}_n)$ can be proved by use of the Borel-Cantelli lemma. $\square$

Appendix B.4. Proof of Theorem 4

We begin by deriving the covering number of $\mathcal{F}_{\text{ReLU}}$ to demonstrate that the assumption in Lemma 8.4 of Van de Geer (2000) (Lemma 3 in Appendix A) holds, which is necessary for proving Theorem 4.

Lemma 6 (covering number of ReLU-IPM). *For any $\epsilon > 0$, we have*

$$\mathcal{N}(\mathcal{F}_{\text{ReLU}}, \|\cdot\|_{1, \mathcal{P}}, \epsilon) \leq \mathcal{O} \left(1 + \frac{1}{\epsilon} \right)^{d+1}.$$

Proof. There exists $\epsilon/2$ -cover V_ϵ of $\mathbb{S}^{d-1}$ in the metric $\|\cdot\|_2$ with cardinality $|V_\epsilon| = \mathcal{O}((1 + 1/\epsilon)^d)$. Hence, for any $\boldsymbol{\theta} \in \mathbb{S}^{d-1}$, we can choose $\tilde{\boldsymbol{\theta}} \in V_\epsilon$ such that $\|\boldsymbol{\theta} - \tilde{\boldsymbol{\theta}}\|_2 \leq \epsilon/2$. In addition, there exists $\epsilon/2$ -cover U_ϵ of $[-1, 1]$ in the metric $\|\cdot\|_2$ with cardinality $|U_\epsilon| = \mathcal{O}(1 + 1/\epsilon)$. Hence, for any $\mu \in [-1, 1]$, we can choose $\tilde{\mu} \in U_\epsilon$ such that $|\mu - \tilde{\mu}| \leq \epsilon/2$. Therefore, letting $\tilde{f}(x) = (\tilde{\boldsymbol{\theta}}^\top \mathbf{x} + \tilde{\mu})_+$, we have, for all $\mathbf{x} \in \mathbb{B}^d$,

$$\begin{aligned}
|f(\mathbf{x}) - \tilde{f}(\mathbf{x})| &= |(\boldsymbol{\theta}^\top \mathbf{x} + \mu)_+ - (\tilde{\boldsymbol{\theta}}^\top \mathbf{x} + \tilde{\mu})_+| \\
&\leq |\boldsymbol{\theta}^\top \mathbf{x} + \mu - (\tilde{\boldsymbol{\theta}}^\top \mathbf{x} + \tilde{\mu})| \quad (\text{by Lips. conti. of ReLU}) \\
&\leq \|\boldsymbol{\theta} - \tilde{\boldsymbol{\theta}}\|_2 \cdot \|\mathbf{x}\|_2 + |\mu - \tilde{\mu}| \quad (\text{by Cauchy-Schwarz ineq.}) \\
&\leq \epsilon.
\end{aligned}$$

□

The following lemma is also needed to prove Theorem 4.

Lemma 7. Let $u(\cdot) := \frac{\pi(\cdot)}{1-\pi(\cdot)}$. For $\{(\mathbf{X}_i, T_i)\}_{i=1}^n$, we define $\tilde{\mathbf{w}} := (\tilde{w}_1, \dots, \tilde{w}_n)^\top \in \mathcal{W}^+$ as

$$\tilde{w}_i := \frac{\mathbb{I}(T_i = 0)u(\mathbf{X}_i)}{\sum_{i:T_i=0} u(\mathbf{X}_i)}, \quad i \in [n]$$

Then, we have

$$d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_0^{\tilde{\mathbf{w}}}, \hat{\mathcal{P}}_1) = \mathcal{O}_p(n^{-1/2}).$$

Proof of Lemma 7. For given $(\mathbf{X}_1, \dots, \mathbf{X}_n) = (\mathbf{x}_1, \dots, \mathbf{x}_n)$, we can write

$$\begin{aligned}
d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_0^{\tilde{\mathbf{w}}}, \hat{\mathcal{P}}_1) &= \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \sum_{i=1}^n \left(\tilde{w}_i - \frac{\mathbb{I}(T_i = 1)}{n_1} \right) f(\mathbf{x}_i) \right| \\
&\leq \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \sum_{i=1}^n \left(\tilde{w}_i - \frac{u(\mathbf{x}_i)\mathbb{I}(T_i = 0)}{n_1} \right) f(\mathbf{x}_i) \right| \quad (\text{B.14})
\end{aligned}$$

$$\begin{aligned}
&+ \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \sum_{i=1}^n \left(\frac{u(\mathbf{x}_i)\mathbb{I}(T_i = 0)}{n_1} - \frac{\mathbb{I}(T_i = 1)}{n_1} \right) f(\mathbf{x}_i) \right|. \quad (\text{B.15})
\end{aligned}$$

For (B.14), we have

$$\begin{aligned}
& \frac{n_1}{n} \times (B.14) \\
&= \frac{\sum_{i=1}^n \mathbb{I}(T_i = 1)}{n} \\
&\quad \times \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \sum_{i=1}^n \left(\frac{\mathbb{I}(T_i = 0)u(\mathbf{x}_i)}{\sum_{j=1}^n \mathbb{I}(T_j = 0)u(\mathbf{x}_j)} - \frac{u(\mathbf{x}_i)\mathbb{I}(T_i = 0)}{\sum_{j=1}^n \mathbb{I}(T_j = 1)} \right) f(\mathbf{x}_i) \right| \\
&= \left| \frac{\sum_{i=1}^n \mathbb{I}(T_i = 1)}{n} - \frac{\sum_{j=1}^n \mathbb{I}(T_j = 0)u(\mathbf{x}_j)}{n} \right| \\
&\quad \times \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \frac{\sum_{i=1}^n \mathbb{I}(T_i = 0)u(\mathbf{x}_i)f(\mathbf{x}_i)}{\sum_{j=1}^n \mathbb{I}(T_j = 0)u(\mathbf{x}_j)} \right| \quad (\text{by Hoeffding's ineq.}) \\
&\leq 2 \left| \frac{\sum_{i=1}^n \mathbb{I}(T_i = 1)}{n} - \frac{\sum_{i=1}^n \mathbb{I}(T_i = 0)u(\mathbf{x}_i)}{n} \right| \\
&= \mathcal{O}_p(n^{-1/2}),
\end{aligned}$$

where $\mathcal{O}_p$ is defined in probability with respect to $\{T_i\}_{i=1}^n$ conditional on $(\mathbf{X}_1, \dots, \mathbf{X}_n) = (\mathbf{x}_1, \dots, \mathbf{x}_n)$. Here, the last inequality holds by the inequality $\|f(z)\|_2 \leq |z^\top \boldsymbol{\theta} + \mu| \leq 2$.

For (B.15), let $W_i := u(\mathbf{x}_i)\mathbb{I}(T_i = 0) - \mathbb{I}(T_i = 1)$. Then, we have $\mathbb{E}(W_i) = 0$ and $\max_{i \in [n]} K^2(\mathbb{E}e^{|W_i|^2/K^2} - 1) \leq \sigma_0^2$, for $K = \eta^{-1}$ and $\sigma_0^2 = K^2(e - 1)$. Hence, by Theorem 3 and Lemma 8.4 of Van de Geer (2000) (see Lemma 3 in Appendix A), there exists a constant $c > 0$ such that

$$\mathbb{P} \left(\sup_{f \in \mathcal{F}_{\text{ReLU}}} \frac{\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i f(\mathbf{x}_i)}{\sqrt{\frac{1}{n} \sum_{i=1}^n f(\mathbf{x}_i)^2}} \geq c^{\frac{3}{2}} \mid (\mathbf{X}_1, \dots, \mathbf{X}_n) = (\mathbf{x}_1, \dots, \mathbf{x}_n) \right) \leq ce^{-c},$$

and hence

$$\mathbb{P} \left(\sup_{f \in \mathcal{F}_{\text{ReLU}}} \frac{1}{\sqrt{n}} \sum_{i=1}^n W_i f(\mathbf{x}_i) \geq 2c^{\frac{3}{2}} \mid (\mathbf{X}_1, \dots, \mathbf{X}_n) = (\mathbf{x}_1, \dots, \mathbf{x}_n) \right) \leq ce^{-c}.$$

Then, we have

$$\begin{aligned} \frac{n_1}{n} \times (B.15) &= \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \sum_{i=1}^n \left(\frac{u(\mathbf{x}_i) \mathbb{I}(T_i = 0)}{n} - \frac{\mathbb{I}(T_i = 1)}{n} \right) f(\mathbf{x}_i) \right| \\ &= \frac{1}{n} \sup_{f \in \mathcal{F}_{\text{ReLU}}} \left| \sum_{i=1}^n W_i f(\mathbf{x}_i) \right| \\ &= \mathcal{O}_p(n^{-1/2}), \end{aligned}$$

where $\mathcal{O}_p$ is defined in probability with respect to $\{T_i\}_{i=1}^n$ conditional on $(\mathbf{X}_1, \dots, \mathbf{X}_n) = (\mathbf{x}_1, \dots, \mathbf{x}_n)$.

To sum up, we get

$$d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_0^{\hat{\mathbf{w}}}, \hat{\mathcal{P}}_1) = \mathcal{O}_p(n^{-1/2})$$

conditional on $(\mathbf{X}_1, \dots, \mathbf{X}_n) = (\mathbf{x}_1, \dots, \mathbf{x}_n)$. Since it holds for every $\{\mathbf{x}_i\}_{i=1}^n \in \mathcal{X}^n$, we obtain the assertion and complete the proof. $\square$

Proof of Theorem 4. We can decompose the error of $\widehat{\text{ATT}}^{\hat{\mathbf{w}}}$ as

$$\begin{aligned} \widehat{\text{ATT}}^{\hat{\mathbf{w}}} - \text{ATT} &= \sum_{i:T_i=1} \frac{Y_i}{n_1} - \sum_{i:T_i=1} \frac{\hat{w}_i Y_i}{n_1} - \text{SATT} + (\text{SATT} - \text{ATT}) \\ &= \sum_{i:T_i=1} \frac{Y_i}{n_1} - \sum_{i:T_i=1} \frac{\hat{w}_i Y_i}{n_1} \end{aligned} \quad (\text{B.16})$$

$$\begin{aligned} &- \frac{1}{n_1} \sum_{i:T_i=1} (m_1(\mathbf{X}_i) - m_0(\mathbf{X}_i)) + (\text{SATT} - \text{ATT}) \\ &= \frac{1}{n_1} \sum_{i:T_i=1} m_0(\mathbf{X}_i) - \sum_{i:T_i=0} \hat{w}_i m_0(\mathbf{X}_i) \end{aligned} \quad (\text{B.17})$$

$$+ \sum_{i:T_i=1} \frac{Y_i - m_1(\mathbf{X}_i)}{n_1} - \sum_{i:T_i=0} \hat{w}_i (Y_i - m_0(\mathbf{X}_i)) \quad (\text{B.18})$$

$$+ (\text{SATT} - \text{ATT}). \quad (\text{B.19})$$

By the definition of $\hat{\mathbf{w}}$ and Lemma 7,

$$d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_0^{\hat{\mathbf{w}}}, \hat{\mathcal{P}}_1) \leq d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_0^{\hat{\mathbf{w}}}, \hat{\mathcal{P}}_1) = \mathcal{O}_p(n^{-1/2}). \quad (\text{B.20})$$

By Theorem 2, from the assumption $m_0 \in \mathcal{H}_d^\beta$, we have (i) if $\beta \geq \frac{d+3}{2}$,

$$(B.17) \leq d_{\mathcal{H}_d^\beta}(\hat{\mathcal{P}}_0^{\hat{\mathbf{w}}}, \hat{\mathcal{P}}_1) \lesssim d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_0^{\hat{\mathbf{w}}}, \hat{\mathcal{P}}_1) = \mathcal{O}_p(n^{-1/2}),$$

and (ii) if $\beta < \frac{d+3}{2}$,

$$(B.17) \leq d_{\mathcal{H}_d^\beta}(\hat{\mathcal{P}}_0^{\hat{\mathbf{w}}}, \hat{\mathcal{P}}_1) \lesssim d_{\mathcal{F}_{\text{ReLU}}}(\hat{\mathcal{P}}_0^{\hat{\mathbf{w}}}, \hat{\mathcal{P}}_1)^{\frac{2\beta}{d+3}} = \mathcal{O}_p(n^{-\frac{\beta}{d+3}}).$$

Since $\{\hat{w}_i\}_{i=1}^n$ is a measurable function of $\{(\mathbf{X}_i, T_i)\}_{i=1}^n$, we have

$$\mathbb{E} \left(\sum_{i:T_i=1} \frac{Y_i - m_1(\mathbf{X}_i)}{n_1} - \sum_{i:T_i=0} \hat{w}_i(Y_i - m_0(\mathbf{X}_i)) \middle| \{(\mathbf{X}_i, T_i)\}_{i=1}^n \right) = 0.$$

and hence

$$\mathbb{E}[(B.18)] = 0.$$

Also,

$$\begin{aligned} & \text{Var}[(B.18)] \\ &= \mathbb{E} \left(\text{Var} \left(\sum_{i:T_i=1} \frac{Y_i - m_1(\mathbf{X}_i)}{n_1} - \sum_{i:T_i=0} \hat{w}_i(Y_i - m_0(\mathbf{X}_i)) \middle| \{(\mathbf{X}_i, T_i)\}_{i=1}^n \right) \right) \\ & \quad + \text{Var} \left(\mathbb{E} \left(\sum_{i:T_i=1} \frac{Y_i - m_1(\mathbf{X}_i)}{n_1} - \sum_{i:T_i=0} \hat{w}_i(Y_i - m_0(\mathbf{X}_i)) \middle| \{(\mathbf{X}_i, T_i)\}_{i=1}^n \right) \right) \\ &= \mathbb{E} \left(\sum_{i:T_i=1} \text{Var} \left(\frac{Y_i(1)}{n_1} \middle| \{(\mathbf{X}_i, T_i)\}_{i=1}^n \right) + \sum_{i:T_i=0} \text{Var}(\hat{w}_i Y_i(0) \middle| \{(\mathbf{X}_i, T_i)\}_{i=1}^n) \right) \\ &= \mathbb{E} \left(\sum_{i:T_i=1} \frac{\text{Var}(Y_i(1) \middle| \{(\mathbf{X}_i, T_i)\}_{i=1}^n)}{n_1^2} + \sum_{i:T_i=0} \hat{w}_i^2 \text{Var}(Y_i(0) \middle| \{(\mathbf{X}_i, T_i)\}_{i=1}^n) \right) \\ &= \mathcal{O}(n^{-1}), \end{aligned}$$

where we use the fact that $\max_{i \in [n]} \hat{w}_i < (n_0 \eta^2)^{-1}$. Hence, by the Chebyshev's inequality, we get

$$(B.18) = \mathcal{O}_p(n^{-1/2}).$$

Since

$$(B.19) = \mathcal{O}_p(n^{-1/2})$$

holds by the Central Limit Theorem, we obtain the assertion of Theorem 4. $\square$

Appendix B.5. Proof of Theorem 5

Proof. For every $g \in \mathcal{H}_d^\beta$,

$$\begin{aligned} \Delta \text{DP}_\phi(g \circ h) &= \left| \mathbb{E}_0(\phi(g \circ h(\mathbf{X}, 0))) - \mathbb{E}_1(\phi(g \circ h(\mathbf{X}, 1))) \right| \\ &\leq C \cdot \left| \mathbb{E}_0(g(h(\mathbf{X}, 0))) - \mathbb{E}_1(g(h(\mathbf{X}, 1))) \right| \\ &\quad \text{(by def. of Lipschitz function)} \\ &\leq C \cdot \sup_{g \in \mathcal{H}_d^\beta} \left| \int g(\mathbf{z})(d\mathcal{P}_0^h(\mathbf{z}) - d\mathcal{P}_1^h(\mathbf{z})) \right|, \end{aligned}$$

where C is the Lipschitz constant of ϕ . By Theorem 2, we have

$$\sup_{g \in \mathcal{H}_d^\beta} \left| \int g(\mathbf{z})(d\mathcal{P}_0^h(\mathbf{z}) - d\mathcal{P}_1^h(\mathbf{z})) \right| \leq c \cdot d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}_0^h, \mathcal{P}_1^h)^{\frac{2\beta}{d+3}},$$

for $\beta < \frac{d+3}{2}$, and

$$\sup_{g \in \mathcal{H}_d^\beta} \left| \int g(\mathbf{z})(d\mathcal{P}_0^h(\mathbf{z}) - d\mathcal{P}_1^h(\mathbf{z})) \right| \leq c \cdot d_{\mathcal{F}_{\text{ReLU}}}(\mathcal{P}_0^h, \mathcal{P}_1^h),$$

for $\beta \geq \frac{d+3}{2}$, where c is some positive constant that only depends on d and β . $\square$

Appendix C. Experimental Details for ReLU-CB

Appendix C.1. Simulation model

We generate simulation datasets under two scenarios, A and B, to evaluate the finite-sample performance of the ReLU-CB estimator, where both the outcome and propensity score models are nonlinear. Let $\mathbf{X} \in \mathbb{R}^d$ be the covariates. For the propensity score, we generate the binary treatment indicators $T \in \{0, 1\}$ from

$$\mathbb{P}(T = 1 \mid \mathbf{X}) = \text{sig}(\tau \cdot g(\mathbf{X})),$$

where $g(\cdot)$ is a model-specific function, and τ controls the strength of treatment assignment and, consequently, the degree of overlap (lower values indicate higher overlap, while higher values indicate lower overlap). For each scenarios,

we consider two variants: A1 and B1 with $\tau = 1$, and A2 and B2 with $\tau = 10$. The potential outcomes for $t \in \{0, 1\}$ are generated as

$$Y(t) = m_t(\cdot) + \varepsilon_t,$$

with $\varepsilon_t \stackrel{\text{ind}}{\sim} \mathcal{N}(0, 1)$.

1. Scenario A (Kang and Schafer, 2007): For each unit $i = 1, \dots, n$, we consider transformations $\mathbf{X}_i = (X_{i1}, X_{i2}, X_{i3}, X_{i4})^\top$ of latent variables $\mathbf{Z}_i = (Z_{i1}, Z_{i2}, Z_{i3}, Z_{i4})^\top$ independently generated from $\mathcal{N}(0, I_4)$, which are given as

$$\begin{aligned} X_{i1} &= \exp(Z_{i1}/2), \\ X_{i2} &= Z_{i1}/(1 + \exp(Z_{i1})) + 10, \\ X_{i3} &= (Z_{i1}Z_{i3}/25 + 0.6)^3, \\ X_{i4} &= (Z_{i2} + Z_{i4} + 20)^2. \end{aligned}$$

Instead of $\{\mathbf{Z}_i\}_{i=1}^n$, only $\{\mathbf{X}_i\}_{i=1}^n$ can be observed. For the propensity score model and the outcome model, we generate them using

$$\begin{aligned} g(\mathbf{Z}_i) &= -Z_{i1} + 0.5Z_{i2} - 0.25Z_{i3} - 0.1Z_{i4}, \\ m_t(\mathbf{Z}_i) &= 210 + 27.4Z_{i1} + 13.7Z_{i2} + 13.7Z_{i3} + 13.7Z_{i4} + \epsilon_i. \end{aligned}$$

The true ATT value of this simulation model is 0.

2. Scenario B (Orihara et al., 2025): We consider $\mathbf{X}_i \in \mathbb{R}^7$ independently generated as $\mathbf{X}_i \sim \mathcal{N}(0, I_7)$. Let

$$\boldsymbol{\alpha} = (0.2, 0.4, -0.125, 0.3, -0.2, -0.4, -0.25, 0.35)^\top,$$

and $\ell(\mathbf{X}) = \boldsymbol{\alpha}^\top (1, \mathbf{X}_1, \dots, \mathbf{X}_7)^\top$. For the propensity score model and the outcome models, we use

$$\begin{aligned} g(\mathbf{X}_i) &= \ell(\mathbf{X}_i) + 0.8 X_{i1}X_{i3} - 0.8 X_{i5}X_{i6} + \sin(2X_{i2}X_{i4}) \\ &\quad + \cos(2X_{i4}X_{i5}) - 0.5 \exp(2X_{i2}X_{i4}) - X_{i2}X_{i5}X_{i6}, \end{aligned}$$

and

$$\begin{aligned} m_0(\mathbf{X}_i) &= 1 + 0.6X_{i1} - 0.5X_{i2}^2 + 0.4X_{i3}X_{i4} + 0.5 \sin(X_{i2}) \\ &\quad + 0.4 \cos(X_{i4}) + 0.3 \exp(0.2X_{i6}), \\ m_1(\mathbf{X}_i) &= m_0(\mathbf{X}_i) + 0.5 + 0.3X_{i1} - 0.2X_{i2}X_{i3} + 0.4 \sin(X_{i4}). \end{aligned}$$

That is, we consider a simulated dataset with heterogeneous treatment effects and estimate the ATT. Using Monte Carlo approximation with 10^5 samples, we approximate the true value of the ATT as 0.526.

Remark 4. *Scenario B follows the large-complexity and poor overlap scenario (setting (f)) in Orihara et al. (2025). While the original data-generating mechanism involves a more complex covariate structure, we adopt a simplified covariate generation scheme by sampling $\mathbf{X}_i \sim \mathcal{N}(0, I_d)$ and assuming full observability of all variables.*

Appendix C.2. Baselines and Implementation Details

We use R (ver. 4.0.2) or Python (ver. 3.9) wherever necessary, and use NVIDIA TITAN Xp GPUs.

- GLM and Boosting: The SIPW estimator is a refined IPW estimator for the ATT using normalized weights such as

$$\sum_{i:T_i=1} \frac{1}{n_1} Y_i - \frac{\sum_{i:T_i=0} \hat{\pi}(\mathbf{X}_i)(1 - \hat{\pi}(\mathbf{X}_i))^{-1} Y_i}{\sum_{i:T_i=0} \hat{\pi}(\mathbf{X}_i)(1 - \hat{\pi}(\mathbf{X}_i))^{-1}},$$

where $\hat{\pi}(\cdot)$ is an estimated propensity score. That is, the weights for the control groups are given by the quantities $\hat{\pi}(\mathbf{X}_i)(1 - \hat{\pi}(\mathbf{X}_i))^{-1} / \{\sum_{i:T_i=0} \hat{\pi}(\mathbf{X}_i)(1 - \hat{\pi}(\mathbf{X}_i))^{-1}\}$. For GLM, we use the linear logistic regression $\hat{\pi}_{\hat{\beta}}(\mathbf{x}) = \{1 + \exp(-\mathbf{x}^\top \hat{\beta})\}^{-1}$, where the regression coefficient $\hat{\beta}$ is estimated by the maximum likelihood estimator using the `GLM` package in R. For Boosting, we use the `twang` package in R (Ridgeway et al., 2022), with 20,000 iterations and a shrinkage parameter set to 0.0005.

- CBPS (Imai and Ratkovic, 2014): Let $\pi_{\beta}(\mathbf{X}_i)$ be a parametric propensity score model, i.e., $\pi_{\beta}(\mathbf{X}_i) := \mathbb{P}(\mathbf{T}_i = 1 | \mathbf{X}_i)$. Imai and Ratkovic (2014) proposes an estimator of $\beta \in \mathbb{R}^d$ based on balancing the moments between the treated and control groups,

$$\frac{1}{n} \sum_{i:T_i=0} \frac{\pi_{\beta}(\mathbf{X}_i)}{1 - \pi_{\beta}(\mathbf{X}_i)} \psi(\mathbf{X}_i) = \frac{1}{n} \sum_{i:T_i=1} \psi(\mathbf{X}_i), \quad (\text{C.1})$$

where $\psi(\cdot) : \mathcal{X} \rightarrow \mathbb{R}^p$ is a pre-specified transformation. That is, they estimate the ATT by

$$\sum_{i:T_i=1} \frac{1}{n_1} Y_i - \frac{\sum_{i:T_i=0} \pi_{\hat{\beta}}(\mathbf{X}_i) \left(1 - \pi_{\hat{\beta}}(\mathbf{X}_i)\right)^{-1} Y_i}{\sum_{i:T_i=0} \pi_{\hat{\beta}}(\mathbf{X}_i) \left(1 - \pi_{\hat{\beta}}(\mathbf{X}_i)\right)^{-1}},$$

where $\hat{\beta}$ is the solution of the equation (C.1). In this study, we use the linear logistic regression model for $\pi_{\beta}(\mathbf{X}_i)$, set $p = d$ and $\psi(\mathbf{x}) = \mathbf{x}$, and implement CBPS using the CBPS package in R (Fong et al., 2022) with default parameters.

- EB (Hainmueller, 2012): Hainmueller (2012) solves

$$\begin{aligned} & \underset{\mathbf{w}}{\text{minimize}} \sum_{i:T_i=0} w_i \log w_i \\ & \text{subject to} \sum_{i:T_i=0} w_i \psi(\mathbf{X}_i) = \frac{1}{n_1} \sum_{i:T_i=1} \psi(\mathbf{X}_i), \quad \sum_{i:T_i=0} w_i = 1, \quad w_i > 0, \end{aligned}$$

where $\mathbf{w} = (w_1, \dots, w_n)^\top$ and $\psi(\cdot) : \mathcal{X} \rightarrow \mathbb{R}^p$ is a pre-specified transformation. That is, they estimate the ATT by

$$\sum_{i:T_i=1} \frac{Y_i}{n_1} - \sum_{i:T_i=0} w_i Y_i.$$

We set $p = d$ and $\psi(\mathbf{x}) = \mathbf{x}$, and implement EB using the EB package in R (Hainmueller and Hainmueller, 2022) with default parameters.

- IPMs: We find $\hat{\mathbf{w}}$ by applying the gradient descent and gradient ascent algorithms iteratively until convergence. To be more specific, suppose that the discriminator class $\mathcal{F}$ is parameterized by $\boldsymbol{\lambda} \in \Lambda$. That is, $\mathcal{F} = \{f(\cdot; \boldsymbol{\lambda}) : \boldsymbol{\lambda} \in \Lambda\}$. Then, we estimate $\hat{\mathbf{w}}$ by minimizing $\ell_n(\boldsymbol{\lambda}_{\mathbf{w}}, \mathbf{w})$ with respect to $\mathbf{w}$, where

$$\ell_n(\boldsymbol{\lambda}, \mathbf{w}) := \left\{ \sum_{i:T_i=1} \frac{f(\mathbf{x}_i; \boldsymbol{\lambda})}{n_1} - \sum_{i:T_i=0} w_i f(\mathbf{x}_i; \boldsymbol{\lambda}) \right\}^2$$

and $\boldsymbol{\lambda}_{\mathbf{w}} = \arg\max_{\boldsymbol{\lambda}} \ell_n(\boldsymbol{\lambda}, \mathbf{w})$. To get $\hat{\mathbf{w}}$, we update $\mathbf{w}$ by applying the gradient descent algorithm to $\ell_n(\boldsymbol{\lambda}, \mathbf{w})$ with $\boldsymbol{\lambda}$ being fixed and then

update λ by applying the gradient ascent algorithm with $\mathbf{w}$ being fixed. At each gradient descent update, we repeat $N_{\text{epo}}^{\text{adv}}$ gradient ascent updates, and we repeat this gradient descent-ascent updates N_{epo} many times. Let us denote the learning rates in the gradient descent and ascent steps as lr and lr^{adv} , respectively. In addition, we use the ensemble technique explained in Section 3.3 with $K = 100$ in the gradient ascent step.

- (1) Wasserstein distance (Kong et al., 2023): The Wasserstein distance is computed based on the methods proposed by Gulrajani et al. (2017) and Arjovsky et al. (2017). Specifically, given points $\mathbf{x}_t$ and $\mathbf{x}_c$ from the treated and control groups respectively, we generate 100 samples of $\mathbf{x}(\lambda) := \lambda \mathbf{x}_c + (1 - \lambda) \mathbf{x}_t$ with $\lambda \sim \text{Unif}(0, 1)$. We then compute the gradient of the discriminator with respect to the interpolated samples and penalize its norm to encourage it to approach 1. We set the regularization parameter to 0.3. For the discriminators, we use a 2-layer neural network with 100 hidden nodes at each layer and Leaky ReLU activation. We set $N_{\text{epo}} = 1000$ and $N_{\text{epo}}^{\text{adv}} = 5$ and use the Adam optimizer with the learning rates $\text{lr} = 0.03$ and $\text{lr}^{\text{adv}} = 0.3$ in the gradient descent and ascent steps, respectively.
- (2) MMD: For $\mathbf{w} = (w_1, \dots, w_n)^\top \in \mathcal{W}^+$, the square of the MMD with a kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is given as

$$\begin{aligned} d_{\mathcal{H}_k}(\hat{\mathcal{P}}_0^{\mathbf{w}}, \hat{\mathcal{P}}_1)^2 &= \sum_{i:T_i=0} \sum_{j:T_j=0} w_i w_j k(\mathbf{X}_i, \mathbf{Y}_j) - 2 \sum_{i:T_i=0} \sum_{j:T_j=1} \frac{w_i}{n_1} k(\mathbf{X}_i, \mathbf{Y}_j) \\ &\quad + \sum_{i:T_i=1} \sum_{j:T_j=1} \frac{1}{n_1^2} k(\mathbf{X}_i, \mathbf{Y}_j). \end{aligned} \quad (\text{C.2})$$

We consider two kernels in the experiments as follows: (i) RBF kernel (Kong et al., 2023), (ii) Sobolev kernel (Wong and Chan, 2017). For (i) the RBF kernel $k_\sigma(\mathbf{x}, \mathbf{y}) = \exp(-\|\mathbf{x} - \mathbf{y}\|_2^2 / \sigma^2)$, we use the Adam optimizer with a learning rate of $\text{lr} = 0.03$ and $N_{\text{epo}} = 1000$ for gradient descent steps, and set $\sigma = 10$. For (ii) the Sobolev kernel

$$k_s(\mathbf{x}, \mathbf{y}) = \prod_{j=1}^d \{1 + k_1(\mathbf{x}_j)k_1(\mathbf{y}_j) + k_2(\mathbf{x}_j)k_2(\mathbf{y}_j) - k_4(|\mathbf{x}_j - \mathbf{y}_j|)\},$$

where $k_1(t) = t - 0.5$, $k_2(t) = \frac{k_1(t)^2 - \frac{1}{12}}{2}$, and $k_4(t) = \frac{k_1(t)^4 - \frac{k_1(t)^2}{2} + \frac{7}{240}}{24}$, we use Adam optimizer with $\text{lr} = 0.03$ and $N_{\text{epo}} = 1000$ for gradient descent steps.

- (3) SIPM (Kong et al., 2023): We set $N_{\text{epo}} = 1000$ and $N_{\text{epo}}^{\text{adv}} = 3$ and use the Adam optimizer with $\text{lr} = 0.1$ for the gradient descent step and the SGD optimizer with $\text{lr}^{\text{adv}} = 1.0$ for the gradient ascent step.
- (4) Hölder-IPM: We use a fully connected neural network with $L = \{1, 2\}$ hidden layers and 100 nodes at each layer. By clipping the output value and the weights, we control the sup-norm of neural networks being less than or equal to 1, and the maximum absolute values of the weights bounded by $\sqrt{d}$, where d is the input dimension. We set $N_{\text{epo}} = 1000$ and $N_{\text{epo}}^{\text{adv}} = 1$ and use the Adam optimizer with $\text{lr} = 0.01$ for the gradient descent step and the SGD optimizer with $\text{lr}^{\text{adv}} = 0.01$ for the gradient ascent step.
- (5) ReLU-IPM: We set $N_{\text{epo}} = 1000$ and $N_{\text{epo}}^{\text{adv}} = 1$ and use the Adam optimizer with $\text{lr} = 0.05$ for the gradient descent step and the SGD optimizer with $\text{lr}^{\text{adv}} = 1.0$ for the gradient ascent step.

Appendix D. Experimental Details for ReLU-FRL

Appendix D.1. Datasets

- ADULT: The Adult income dataset (Dua and Graff, 2017) can be downloaded from the UCI repository². We follow the pre-processing of the implementation of IBM’s AIF360 (Bellamy et al., 2018)³.
- DUTCH: The Dutch census dataset can be downloaded from the public GitHub of Quy et al. (2022)⁴. We follow the pre-processing of Quy et al. (2022) in its official GitHub⁵.

²<https://archive.ics.uci.edu/ml/datasets/adult>

³<https://aif360.readthedocs.io/en/stable/>

⁴https://github.com/tailequy/fairness_dataset/tree/main/experiments/data/dutch.csv

⁵https://github.com/tailequy/fairness_dataset/tree/main/experiments/data/

- CRIME: The communities and crime dataset can be downloaded from the UCI repository⁶. We follow the pre-processing of Jiang et al. (2022) in its official GitHub⁷.
- CELEBA: The CelebAMask-HQ is a face image dataset consisting of 30,000 images, which can be downloaded from the public GitHub of Liu et al. (2015)⁸. All images are resized to a fixed resolution of 224×224 , and a pretrained ResNet18 is used to extract 512-dimensional feature embeddings.

Table D.3 below presents the basic information of the four datasets used in our experiments.

Table D.3: The description of the datasets: ADULT, DUTCH, CRIME and CELEBA.

Dataset	ADULT	DUTCH	CRIME	CELEBA
$\mathbf{X}$	Demographic attributes	Demographic attributes	Demographic attributes	Image features (ResNet-18 embeddings)
S	Gender	Gender	% of black population (binarized)	Gender (Male)
Y	Outcome over \$50k	High-level occupation	% of crimes (binarized)	Smiling
d	101	58	121	512
Train data size	30,136	48,336	1,794	24,000
Test data size	15,086	12,084	200	6,000
Batch size	1024	1024	256	128

Appendix D.2. Implementation Details

Baseline algorithms. For given dataset $\{(\mathbf{x}_i, s_i, y_i)\}_{i=1}^n$, the training objective is to minimize

$$\frac{1}{n} \sum_{i=1}^n \ell(g \circ h(\mathbf{x}_i, s_i), y_i) \quad \text{subject to} \quad d(\mathcal{P}_{0,n}^h, \mathcal{P}_{1,n}^h) \leq \delta,$$

with respect to $g \in \mathcal{G}$ and $h \in \mathcal{H}$, where ℓ is a given task-specific loss function and d is a given deviance measure between the empirical distributions $\mathcal{P}_{0,n}^h$ and $\mathcal{P}_{1,n}^h$ of $\{h(\mathbf{x}_i, 0) : s_i = 0\}$ and $\{h(\mathbf{x}_i, 1) : s_i = 1\}$, respectively. We use the cross-entropy loss for the ℓ . The Lagrangian form of this constrained

⁶<https://archive.ics.uci.edu/dataset/183/communities+and+crime>

⁷https://github.com/zhimengj0326/GDP/blob/master/tabular/data_load.py

⁸<https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html>

optimization can be expressed as

$$\frac{1}{n} \sum_{i=1}^n \ell(g \circ h(\mathbf{x}_i, s_i), y_i) + \lambda d(\mathcal{P}_{0,n}^h, \mathcal{P}_{1,n}^h), \quad (\text{D.1})$$

for some $\lambda > 0$. In our experimental results, we present the Pareto-front trade-off graphs illustrating the relationship between the fairness levels ($\Delta\overline{\text{DP}}, \Delta\text{DP}, \Delta\text{SDP}$) and the binary classification accuracy (ACC) by varying the values of λ .

For the deviance measure, along with the ReLU-IPM for ReLU-FRL, we consider SIPM (Kim et al., 2022a) for SIPM-LFR, Hölder-IPM (Wang et al., 2023) approximating Hölder discriminators by one hidden layer neural networks for Hölder-FRL, MMD-B-Fair (Deka and Sutherland, 2023) and the Jensen-Shannon divergence (LAFTR, Madras et al. (2018)). For LAFTR, we use a single-layered MLP with hidden size 100 and ReLU activation.

Training hyperparameters. We train all models for a total of 200 epochs using the Adam optimizer (Kingma and Ba, 2014). The learning rate is initialized by 0.001 and scheduled by multiplying 0.95 at each epoch. For the optimization we update the discriminator for 2 iterations of the gradient ascent per a single update of the model parameters (i.e. parameters in the representation) by the gradient descent. The learning rate is set to 0.001 and not scheduled. For ReLU-IPM and SIPM, we use the ensemble technique explained in Section 3.3 with $K = 100$ in the gradient ascent step. The reported results are assessed on the test dataset with the final models trained after the 200 epochs. For Hölder-IPM, we use 100 nodes at the hidden layer.

Appendix D.3. Results of Additional Experiments

Figures D.3 and D.4 represent the Pareto-front trade-off graphs of ΔDP and ΔSDP for comparing the ReLU-FRL with the linear logistic prediction head to other baselines. Figure D.5 to Figure D.7 draw the Pareto-front trade-off graphs for the single-layered neural network prediction head. The results are similar to those in Figure 1.

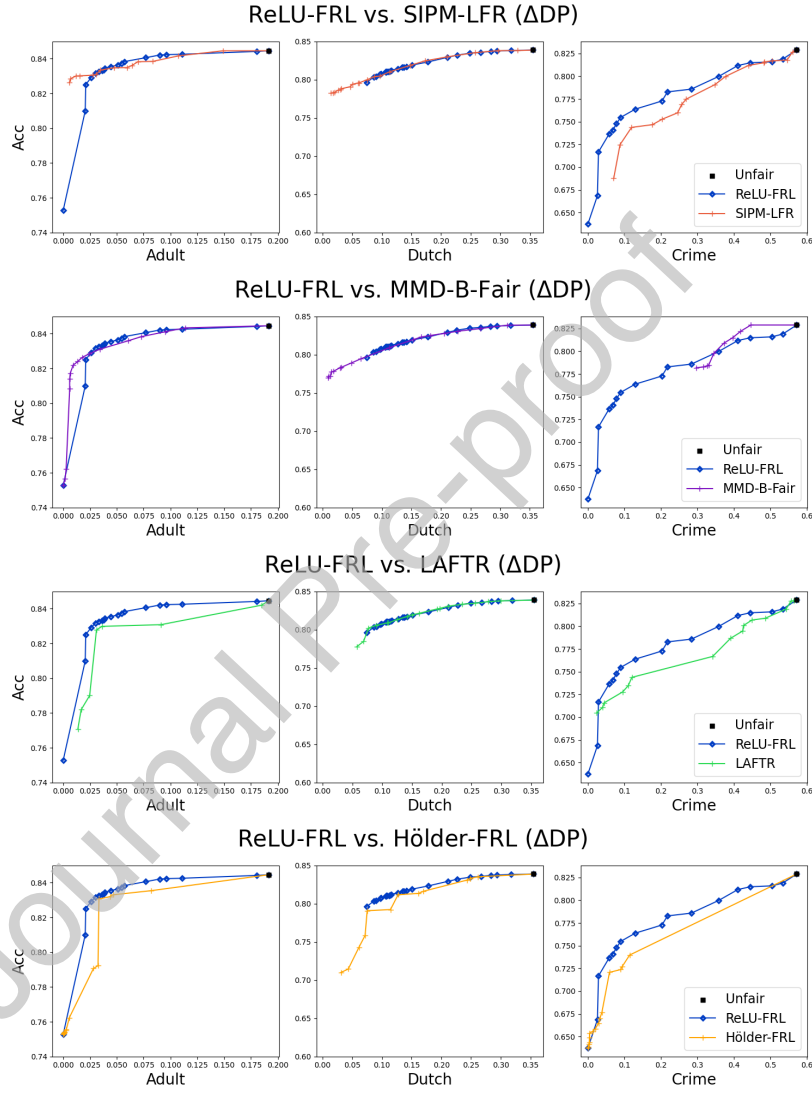


Figure D.3: **Linear logistic prediction head:** Pareto-front lines of fairness level ΔDP and ACC. (Left) ADULT, (Center) DUTCH, (Right) CRIME.

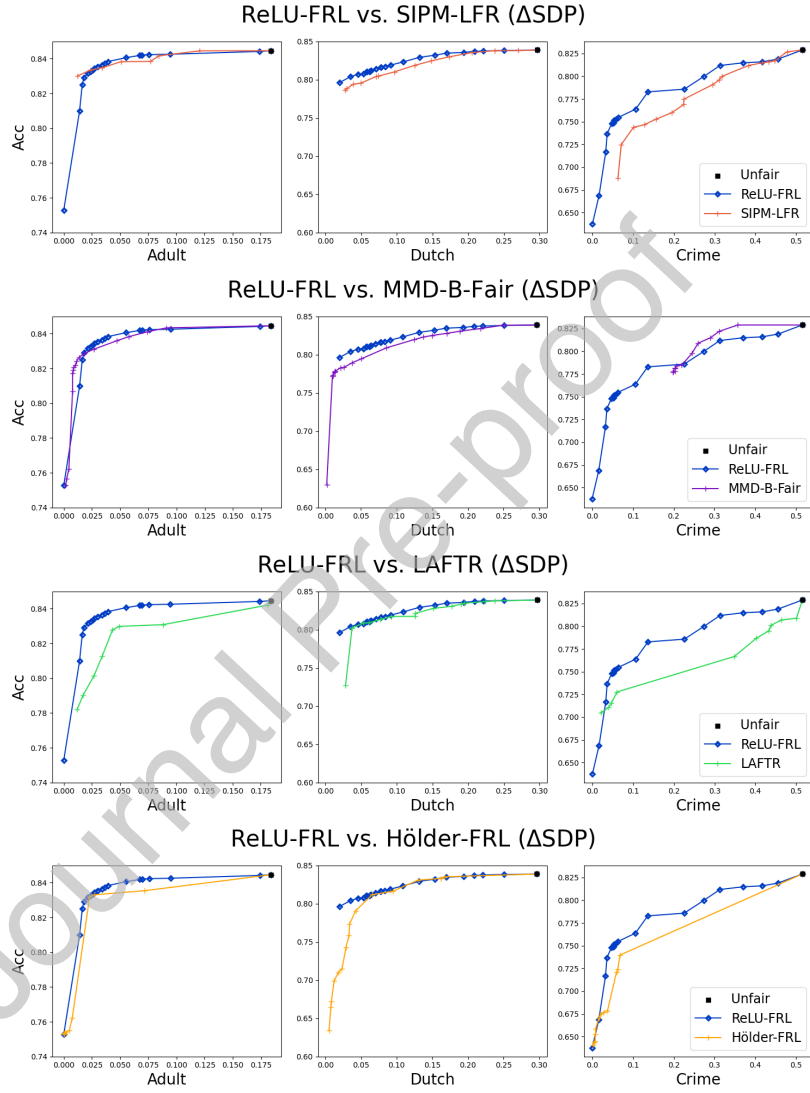


Figure D.4: **Linear logistic prediction head:** Pareto-front lines of fairness level Δ SDP and ACC. (Left) ADULT, (Center) DUTCH, (Right) CRIME.

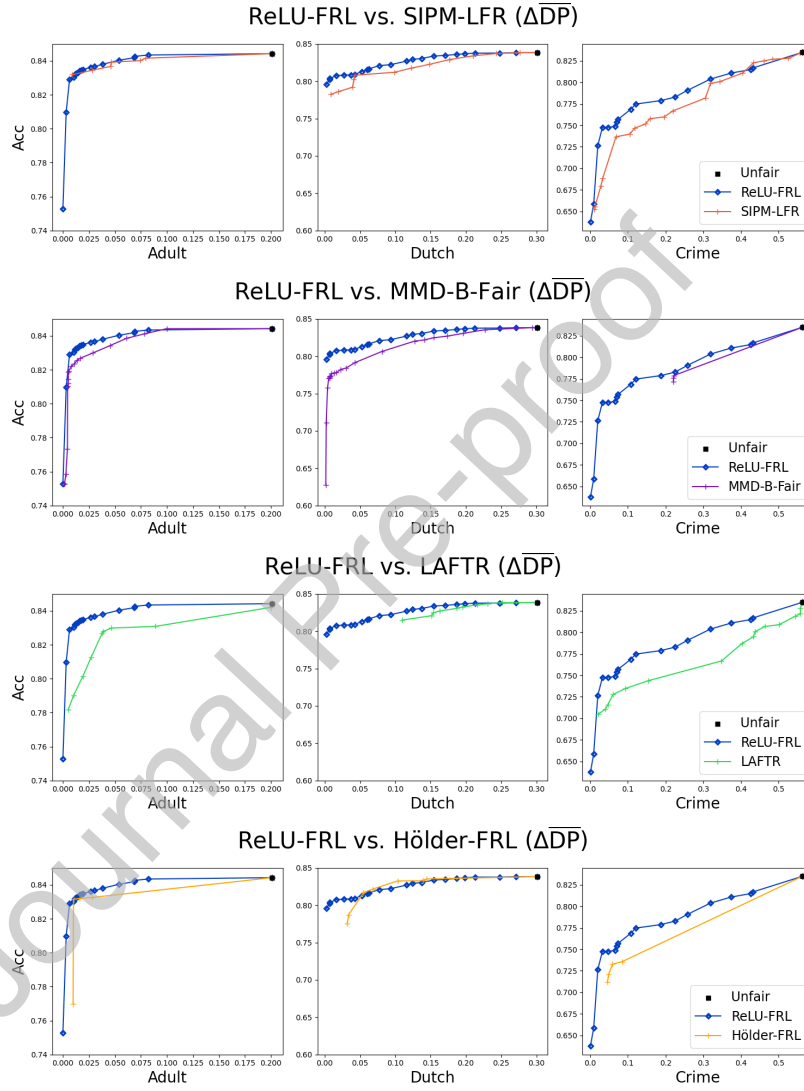


Figure D.5: **Single-layered NN prediction head:** Pareto-front lines of fairness level $\Delta\overline{DP}$ and ACC. (Left) ADULT, (Center) DUTCH, (Right) CRIME.

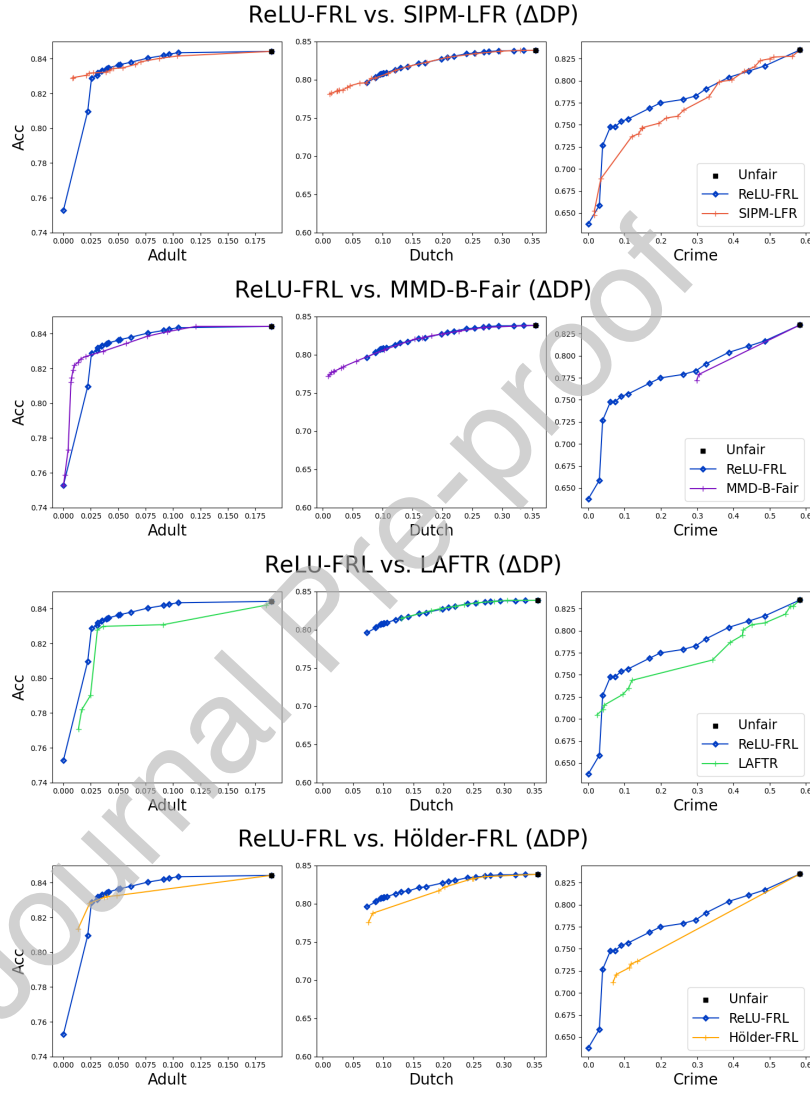


Figure D.6: **Single-layered NN prediction head:** Pareto-front lines of fairness level ΔDP and ACC. (Left) ADULT, (Center) DUTCH, (Right) CRIME.

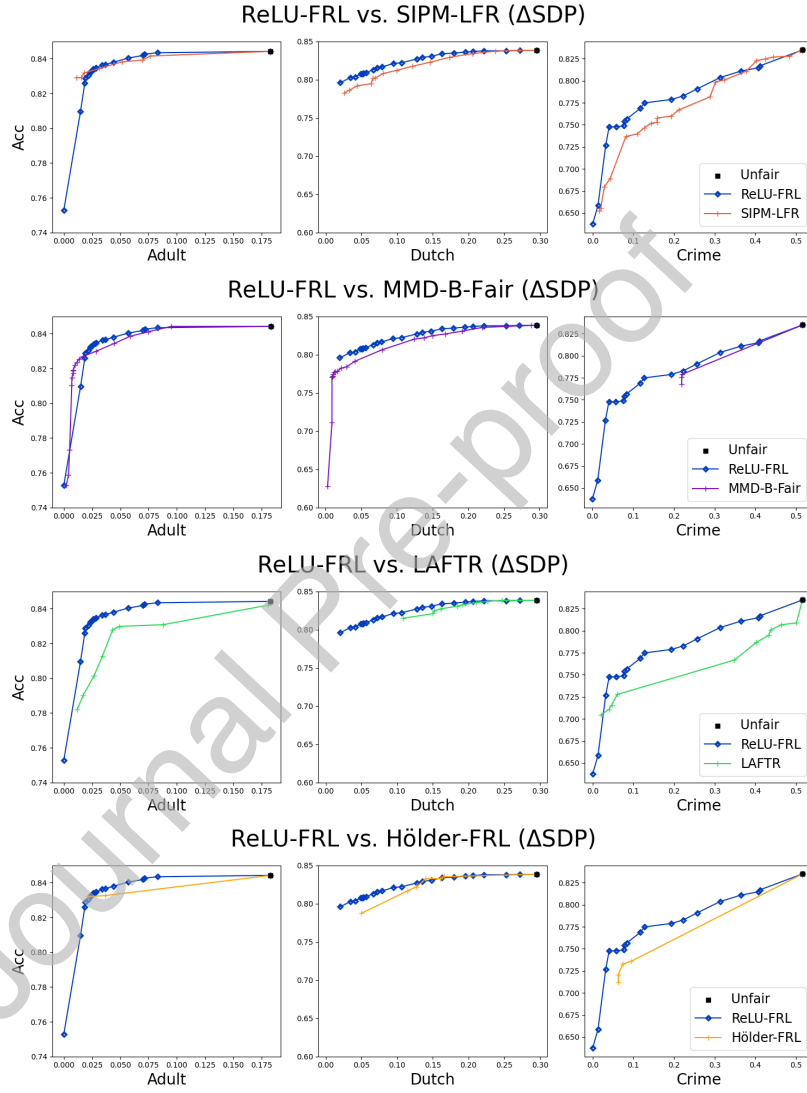


Figure D.7: **Single-layered NN prediction head:** Pareto-front lines of fairness level Δ SDP and ACC. (Left) ADULT, (Center) DUTCH, (Right) CRIME.

Appendix E. Effect of multiple random initializations

To illustrate the effect of multiple random initializations on the computational cost and performance of the proposed method, we evaluate the prediction accuracy (measured by RMSE) and computing time (in seconds) of ReLU-CB. We consider different numbers of initializations $K \in \{1, 10, 100\}$ and sample sizes $n \in \{200, 1000\}$, for simulation setting A1 considered in Section 4.2. The results are presented in Figure E.8. The technique significantly improves accuracy without substantially increasing the computing time.

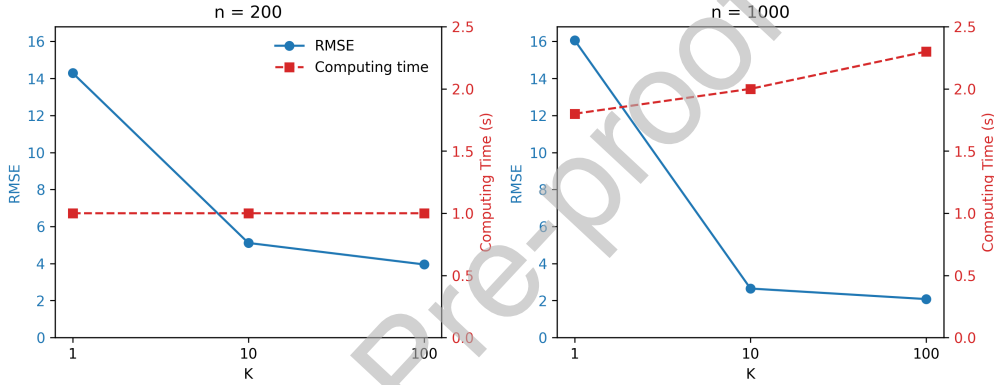


Figure E.8: RMSE and computing time (in seconds) of ReLU-CB for different values of K . The left and right vertical axes correspond to RMSE and computing time, respectively, for Setting A considered in Section 4.2.

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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